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Loss of schizophrenia risk gene *XPO7* disrupts neuronal excitability and network regularity via altered Na^+ channel dynamics in human neurons

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Schizophrenia is a highly heritable psychiatric disorder, yet the molecular mechanisms by which genetic risk contributes to disease pathophysiology remain largely unknown. In this study, we investigate the functional consequences of *XPO7* loss of function (LoF) in human induced pluripotent stem cell (iPSC)-derived neurons, focusing on its role as a schizophrenia risk gene identified through recent large-scale exome sequencing analyses. By integrating high-precision electrophysiological measurements with transcriptomic, proteomic, and imaging approaches, we demonstrate that *XPO7* LoF alters Na^+ channel properties and availability, disrupts neuronal excitability, and impairs the synchrony and regularity of network activity. These functional deficits are accompanied by widespread molecular dysregulation affecting nucleocytoplasmic transport, ion channel function, and synaptic composition. Among the dysregulated proteins is $Na_v1.2$, a voltage-gated sodium channel encoded by *SCN2A*, which displays aberrant subcellular distribution in *XPO7* LoF neurons. Together, these findings position *XPO7* as a critical regulator of neuronal excitability and connectivity, linking channelopathy to cellular phenotypes relevant to schizophrenia pathophysiology.

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INTRODUCTION

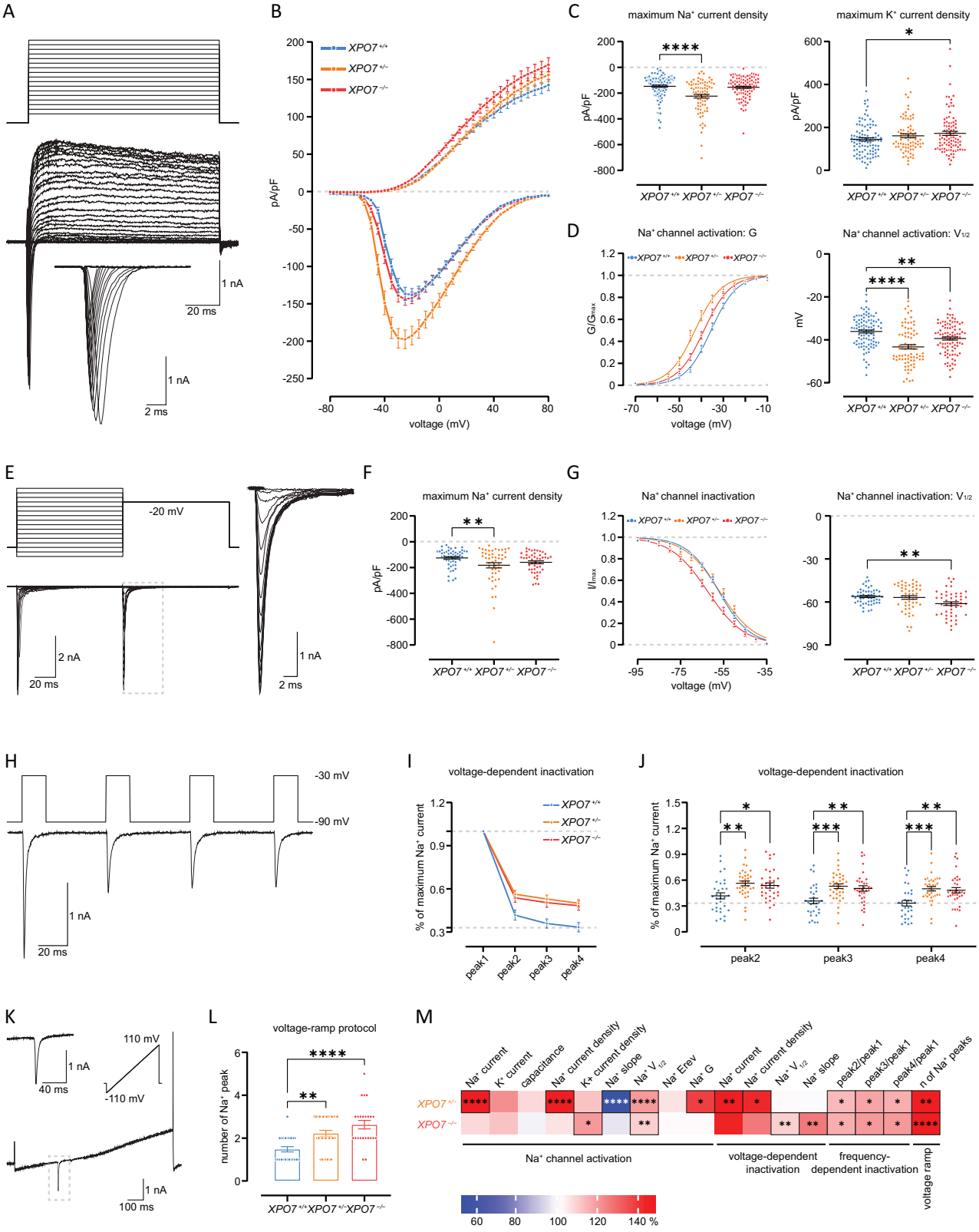
Schizophrenia is a chronic and highly disabling mental disorder with high heritability [1–4]. It affects approximately 1% of the population, yet effective long-term treatments are lacking. Despite its substantial heritability, the molecular mechanisms underlying schizophrenia pathophysiology remain poorly understood. Large-scale genome-wide association studies (GWAS) have identified hundreds of genomic loci associated with schizophrenia susceptibility [5, 6]. More recently, exome sequencing studies have uncovered genes whose rare LoF coding variants significantly increase schizophrenia risk, with effect sizes much larger than those of common GWAS variants [7]. These high-impact mutations often lead to premature protein truncation, and modeling their LoF may provide insights into the biological mechanisms driving the disease risk. One such gene is *XPO7*, which encodes Exportin-7, a nuclear export receptor with broad substrate specificity [8]. *XPO7* plays a critical role in nucleocytoplasmic transport, a fundamental process essential for maintaining cellular homeostasis.

XPO7 has been implicated in various biological processes, including erythroid nuclear maturation, tumor suppression, and oncogenesis [9–12]. While *XPO7* is widely expressed across all regions of the brain, as indicated by Genotype-Tissue Expression (GTEx) data [13], its specific neuronal functions remain largely uncharacterized. One report showed that *XPO7* regulates Hedgehog signaling by exporting Gli2 from the nucleus, a pathway essential for neural development and differentiation [14]. A recent study demonstrated that *XPO7* haploinsufficiency in mice results in cognitive and social behavioral deficits, alongside disruptions in the nuclear transport of molecules associated with schizophrenia genetics [15]. These findings suggest that XPO7 plays an important role in maintaining normal neuronal function.

To investigate the impact of *XPO7* LoF on the neurobiology of human neurons, we utilized human iPSC-derived neurons to model disease risk in vitro [16, 17]. iPSC-derived neurons have been employed to explore both common and rare genetic risk factors for schizophrenia. Studies using neurons derived from schizophrenia-discordant twins have reported reduced morphological complexity, increased excitability, and decreased synaptic activity, accompanied by dysregulated expression of synapse-related genes [18, 19]. In another study leveraging a cohort of iPSC-derived neurons from individuals with high schizophrenia polygenic risk scores (PRSs), elevated Na^+ channel activity was observed and found to correlate with clinical symptoms and cognitive performance [20]. Recent investigations have also examined the effects of rare, high-impact genetic variants. For example, human excitatory neurons with *SETD1A* LoF, implicated by schizophrenia genetics, exhibit dysregulated synaptic function and irregular network activity [21, 22]. Similarly, schizophrenia-associated *NRXN1* heterozygous deletions induce synaptic dysfunction and disrupt neuronal network synchrony [23]. Moreover, LoF of *SCN2A*, a Na^+ channel gene associated with both schizophrenia and autism spectrum disorders (ASD), impairs excitatory and inhibitory neurogenesis and leads to

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abnormal neuronal network activity [24, 25]. Together, these findings highlight the utility of iPSC-derived neurons for modeling disease-associated phenotypes and potential converging and diverging cellular phenotypes in human neurons associated with schizophrenia risk.

Building on this framework, we generated isogenic human iPSC lines with complete (*XPO7*$^{-/-}$) and partial (*XPO7*$^{+/-}$) deletions using CRISPR-Cas9 to directly examine the role of *XPO7* in neuronal function and schizophrenia risk. Using a combination of techniques, including proteomics, transcriptomics, and immunofluorescence,

Fig. 1 *XPO7* LoF iNs exhibit distinct electrophysiological properties with elevated Na^+ channel conductance compared to parental controls. **A** Representative traces of whole-cell voltage-gated Na^+ and K^+ currents in response to voltage steps recorded from iNs at day 56 using the automated patch-clamp instrument, SyncroPatch384. **B** Quantification of current-voltage (I/V) relationships for K^+ (top) and Na^+ (bottom) currents in iNs. **C** Quantification of maximum Na^+ (left) and K^+ (right) channel current densities in $XPO7^{+/+}$ ($n = 97$), $XPO7^{+/-}$ ($n = 79$), and $XPO7^{-/-}$ ($n = 93$) neurons. **D** Summary data showing a significant hyperpolarizing shift in Na^+ channel activation in *XPO7* LoF ($XPO7^{+/-}$ and $XPO7^{-/-}$) neurons compared to parental controls. **E** Representative traces of whole-cell voltage-gated Na^+ currents in response to a steady-state inactivation protocol used to measure channel inactivation. **F** Quantification of maximum Na^+ inactivation current densities in $XPO7^{+/+}$ ($n = 54$), $XPO7^{+/-}$ ($n = 49$), and $XPO7^{-/-}$ ($n = 47$) neurons. **G** Summary data showing a significant hyperpolarizing shift in Na^+ channel inactivation in $XPO7^{-/-}$ neurons compared to parental controls. **H** Representative traces of Na^+ currents in response to four consecutive voltage steps. **I, J** Quantification of frequency-dependent inactivation of Na^+ channels in $XPO7^{+/+}$ ($n = 31$), $XPO7^{+/-}$ ($n = 36$), and $XPO7^{-/-}$ ($n = 34$) neurons. Na^+ currents at each voltage step were normalized to the currents generated by the first voltage step (**I**). Summary data showing a significant increase in Na^+ current inactivation at each voltage step in XPO7 LoF ($XPO7^{+/-}$ and $XPO7^{-/-}$) neurons compared to parental controls (**J**). **K** Representative traces of Na^+ currents in response to a voltage-ramp protocol. **L** Quantification of the number of Na^+ peaks in $XPO7^{+/+}$ ($n = 23$), $XPO7^{+/-}$ ($n = 32$), and $XPO7^{-/-}$ ($n = 24$) neurons. **M** Heatmap illustrating key measurements from electrophysiological assessments. Stroked cells indicate significantly altered measurements in the corresponding *XPO7* LoF neurons ($XPO7^{+/-}$ or $XPO7^{-/-}$) compared to parental controls ($XPO7^{+/+}$). All statistical analyses were performed using ANOVA with post hoc Dunnett's multiple comparisons test. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Error bars represent SEM. See also Fig. S1 and Fig. S2.

together with physiological assessments through whole-cell patch-clamp recordings and high-density micro-electrode array (HD-MEA) recordings, we comprehensively analyzed the molecular, cellular, structural, and functional properties of these neurons at both the single-cell and network levels across developmental stages. We demonstrated that *XPO7* LoF leads to: (1) increased Na^+ channel conductance with altered activation and inactivation properties; (2) impaired neuronal excitability and disrupted action potential dynamics; (3) transcriptional changes, particularly affecting Na^+ channel expression; (4) altered neuronal morphology and subcellular Na^+ channel localization; (5) reduced synapse density and altered synaptic proteomic profiles; and (6) disruptions in neuronal network synchrony. Together, our findings establish increased Na^+ channel signaling as a key downstream consequence of *XPO7* LoF that may underlie schizophrenia risk, underscoring its potential relevance for therapeutic development.

RESULTS

XPO7 LoF Alters Electrophysiological Properties in Human Neurons

Human glutamatergic neurons derived from iPSCs of schizophrenia patients have been shown to exhibit notable electrophysiological differences compared to those from healthy controls, including altered voltage-gated Na^+ currents, although the relevance of these findings to disease pathophysiology remains unclear [16, 17, 20, 26]. To directly examine the functional impact of schizophrenia-associated mutations in a controlled genetic background, we established iPSC lines lacking one ($XPO7^{+/-}$) or both ($XPO7^{-/-}$) copies of *XPO7*, and assessed whether *XPO7* LoF affects ionic conductance and neuronal excitability in glutamatergic neurons derived through forebrain patterning and NGN2 induction [27, 28]. This strategy was motivated by exome sequencing data from the SCHEMA consortium [7], which identified rare protein-truncating variants (PTVs) in *XPO7*, including frameshift, stop-gained, and splice site mutations that span both early (e.g., p.Gln17HisfsTer28) and late (e.g., p.Glu1030GlyfsTer11) coding regions, suggesting that haploinsufficiency may underlie disease risk. To enable consistent neuronal differentiation, a doxycycline-inducible NGN2 cassette was integrated into the AAVS1 safe-harbor locus of human iPSCs using TALENs [27], allowing rapid and efficient generation of excitatory glutamatergic neurons.

We subsequently characterized the electrophysiological properties of these neurons. We first optimized whole-cell patch-clamp recordings using the SyncroPatch 384PE to assess Na^+ and K^+ channel conductance in induced neurons (iNs) at day 56 post-differentiation (see Methods). The SyncroPatch 384PE enables GigaΩ seal formation and precise voltage control across 384 cells simultaneously, as previously described [29]. Using a voltage-clamp protocol (Fig. 1A), we measured and analyzed the properties of voltage-gated Na^+ and K^+ conductance. Both heterozygous ($XPO7^{+/-}$) and homozygous ($XPO7^{-/-}$) LoF resulted in altered Na^+ and K^+ channel current densities in day 56 neurons, corresponding to a developmentally advanced stage of in vitro differentiation (Fig. 1B). Specifically, voltage-gated Na^+ channel current density was significantly increased in $XPO7^{+/-}$ neurons, while voltage-dependent K^+ channel current density exhibited a modest but statistically significant increase in $XPO7^{-/-}$ neurons (Fig. 1C). In addition, both *XPO7* LoF neuron types displayed a significant hyperpolarizing shift in the voltage dependence of Na^+ channel activation (Fig. 1D), with a more pronounced shift observed in $XPO7^{+/-}$ neurons. These findings suggest that *XPO7* LoF modifies the properties of voltage-gated Na^+ channels, with a more pronounced effect in the heterozygous condition.

Using a different electrophysiological recording protocol (Fig. 1E and 1F), we observed a significant hyperpolarizing shift in the voltage dependence of Na^+ channel inactivation in $XPO7^{-/-}$ neurons (Fig. 1G). To investigate frequency-dependent inactivation, we applied four 16-ms voltage depolarizations at 40-ms intervals from -90 to -30 mV (Fig. 1H). Both heterozygous and homozygous *XPO7* LoF neurons exhibited significantly reduced inhibition upon repeated depolarization compared to parental control neurons, suggesting that a greater proportion of Na^+ channels remain available for activation following repeated depolarizations in *XPO7* LoF neurons (Fig. 1I and 1J). These electrophysiological observations were replicated across two independent differentiation batches and confirmed in multiple patch-clamp experiments for each batch (Fig. S1A–D and S2A–D).

Next, we performed manual patch-clamp recordings and applied a voltage ramp protocol to each neuron, quantifying the number of inward Na^+ current peaks as a proxy for neuronal excitability (Fig. 1K), as previously described [20]. Both $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons exhibited an increased number of Na^+ peaks (Fig. 1L–M, and Fig. S2F), suggesting enhanced excitability as a result of *XPO7* LoF.

Using electrophysiological approaches, we comprehensively characterized the properties of neurons under *XPO7* LoF, including Na^+ channel availability during voltage-dependent activation, inactivation, and frequency-dependent inactivation, as well as the number of Na^+ peaks evoked by a voltage ramp (Fig. 1M). Across multiple voltage-clamp protocols, we observed consistent and significant alterations indicating increased Na^+ channel conductance and availability. These findings demonstrate that *XPO7* LoF enhances Na^+ channel activity in human neurons and highlight a key role for *XPO7* in maintaining normal electrophysiological function. These electrophysiological features resemble those observed in neurons derived from individuals with high polygenic

risk scores for schizophrenia [20]. This convergence suggests that both common and rare genetic risk factors may disrupt shared ion channel regulatory pathways, contributing to neuronal hyperexcitability in schizophrenia.

XPO7 LoF Disrupts Neuronal Excitability and Action Potential Dynamics

To further examine the role of *XPO7* in modulating neuronal excitability, we analyzed action potential properties and neuronal activity using a high-density microelectrode array (HD-MEA) system. This platform contains 26,400 microelectrodes within a 3.85 × 2.10 mm^2 sensing area, enabling large-scale ensemble recordings of neuronal electrical activity at high temporal and spatial resolution over developmental periods [30]. Our data processing pipeline for extracting single-neuron electrophysiological features was established in previous studies [31–33]. Raw recordings were filtered and spike-sorted to identify individual units, referring to putative single neurons distinguished by their unique spike waveform characteristics, using previously published methods (Fig. 2A). From the sorted spikes, we extracted single-cell activity metrics, including spike amplitude, firing rate, and interspike interval (ISI), as well as waveform metrics that reflect the signal characteristics of each neuron's action potential. The HD-MEA system also enabled tracking of action potential propagation from the axon initial segment along the axonal arbor and quantification of propagation dynamics at subcellular resolution. Neuronal activity was recorded from *XPO7* LoF and parental control neurons across developmental stages in five independent differentiations, with weekly measurements from day 28 to day 56 post-neuronal induction. Principal component analysis (PCA) of extracted excitability features revealed genotype-specific clustering that was consistently observed across biological replicates (Fig. 2B). A graded, dosage-dependent separation across genotypes indicated that our analysis revealed distinct excitability profiles associated with *XPO7* LoF. To exclude clone-specific effects, we used two independent clones for each *XPO7* LoF line. PCA of recordings from each clone consistently demonstrated genotype-dependent differences in neuronal excitability (Fig. S4G). Unless otherwise noted, sample sizes are reported as n = X/Y, where X indicates the number of HD-MEA cultures analyzed per genotype and Y refers to the number of independent neuronal differentiation batches.

Analysis of HD-MEA recordings revealed that *XPO7* LoF significantly altered the dynamics and waveform characteristics of action potentials (Fig. 2C–E). Consistent with changes in Na^+ channel properties, *XPO7* LoF neurons exhibited elevated spike amplitudes (Fig. 2D) and steeper repolarization and recovery slopes across multiple stages of neuronal differentiation (Fig. 2E and Fig. S4A–F). Both LoF lines displayed shortened peak-to-valley durations, with a more pronounced reduction in $XPO7^{-/-}$ neurons, indicating accelerated repolarization dynamics (Fig. 2E and Fig. S4E). Notably, the features more strongly affected in $XPO7^{+/-}$ neurons included elevated firing rates and reduced mean interspike intervals (Fig. 2D and Fig. S3A). These observations are consistent with our patch-clamp findings (Fig. 1C), which showed a greater increase in Na^+ current density in $XPO7^{+/-}$ compared to $XPO7^{-/-}$ neurons. Importantly, most electrophysiological differences between *XPO7* LoF and control neurons were detectable at all assessed time points, underscoring the persistent phenotype throughout differentiation.

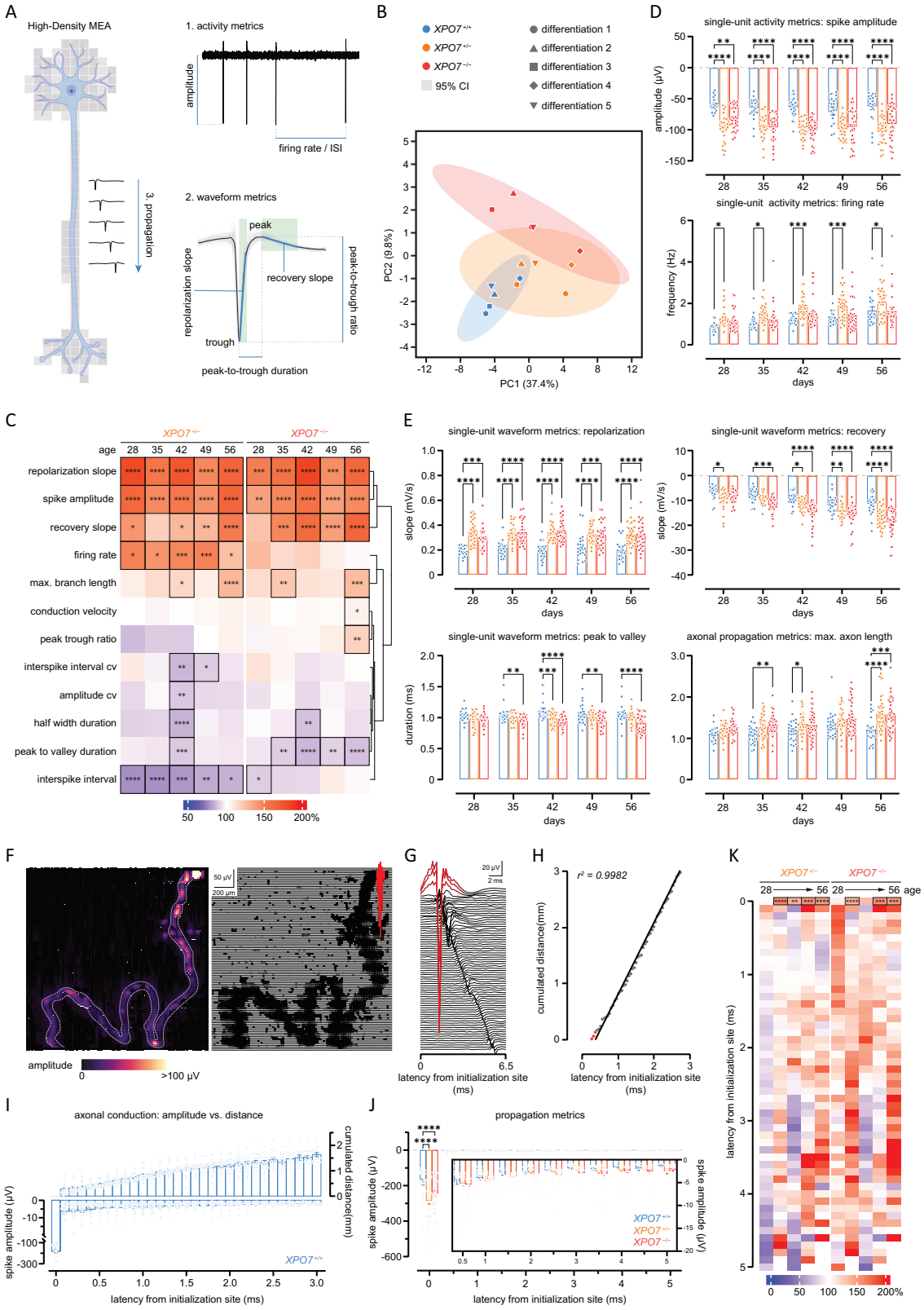
To investigate action potential propagation and subcellular electrical dynamics, we computed array-wide spike-triggered averaged signals for each neuron and reconstructed their spatiotemporal distributions (Fig. 2F). This approach enabled identification and localization of the axon initial segment (AIS), indicated by the largest action potential amplitude and earliest onset, and allowed for tracking of action potential propagation from the AIS to the axonal terminals. Using the HD-MEA system, we quantified propagation velocities by measuring the movement of action potentials in space (cumulative distance) and time (latency from the initiation site) across multiple neighboring electrodes positioned along individual axons (Fig. 2G) and validated this method by demonstrating a strong correlation between distance and latency (Fig. 2H). Given the pronounced increase in action potential amplitudes and elevated Na^+ channel availability at the soma, we asked whether these electrophysiological alterations extended to the distal axon. Characterization of control neurons at day 42 showed action potentials, initiated at the AIS, exhibited the largest extracellular voltage signals within propagation sequences and propagated along the axon with a gradual decline in amplitude (Fig. 2I). This attenuation reflects intrinsic geometric and biophysical properties of the axons, including narrower diameter and lower ion channel density in the distal axon relative to the AIS. Across neuronal maturation, the effect of *XPO7* LoF on increasing action potential amplitude was most prominent at the initiation site (Fig. 2J-K and Fig. S3G-J). Analysis of propagation metrics revealed that conduction velocity in *XPO7* LoF neurons remained largely unchanged across developmental stages, with a modest but statistically significant increase observed at later maturation stages (day 56; Fig. S3F). Measurements of maximum axonal length also revealed an increase under *XPO7* LoF, most evident in day 56 neurons (Fig. 2E).

These findings indicate that *XPO7* plays an important role in maintaining neuronal excitability, with *XPO7* LoF leading to elevated action potential amplitudes and disrupted waveform dynamics. These effects were most prominent at the soma and its vicinity, with relatively limited impact in the distal axon.

XPO7 LoF Reshapes Transcriptomic Profiles

To dissect the mechanisms by which *XPO7* regulates ion channel activity and neuronal excitability, we analyzed the transcriptional profiles of neurons across developmental stages. Given that our results revealed robust changes in neuronal excitability starting at day 28 of differentiation, most of which were maintained or further progressed by day 56, we focused on three developmental time points: day 28, day 42, and day 56, in both *XPO7* LoF neurons and parental controls.

PCA of bulk RNA sequencing data from control neurons showed clustering of biological replicates by age, reflecting both the reproducibility at each time point and divergence across different developmental stages (Fig. 3A). Hierarchical clustering of gene expression profiles further highlighted dynamic transcriptional shifts during neuronal maturation (Fig. 3B), with distinct pathways becoming either activated or repressed throughout iPSC-derived neuron differentiation (Fig. 3C). Differentially expressed genes (DEGs) were hierarchically clustered using fuzzy c-means clustering [34, 35] based on their expression profiles across the three differentiation time points, and three distinct trajectories of upregulated gene expression were identified: (1) progressive upregulation across each time point, indicating a gradual transition in cell state throughout maturation (cluster "UP1"); (2) a significant increase from day 28 to day 42, followed by a plateau (cluster "UP2"); and (3) minimal changes from day 28 to day 42, followed by significant upregulation (cluster "UP3"), suggesting a late-onset expression pattern during maturation. Consistent with these temporal trends, *XPO7* showed a moderate and progressive upregulation during neuronal differentiation (day 42 vs. day 28, $\log_2$ fold-change = 0.13; day 56 vs day 42, $\log_2$ fold-change = 0.23). Gene Ontology (GO) enrichment analysis revealed significant enrichment of genes associated with neuronal development and maturation within the "UP1" and "UP2" groups, including those involved in synaptic function, neuronal morphogenesis, ion channel physiology, and the regulation of neuronal excitability (Fig. 3C). A similar trend was observed for genes linked to neurological disorders (DisGeNET) [36], including schizophrenia, bipolar disorder, and neurodevelopmental disorders, suggesting that genes



associated with disease-relevant symptoms may become active during maturation in iPSC-derived neuron models. Notably, genes encoding voltage-gated Na^+ channels showed a stronger association with the "UP1" group than with "UP2," indicating that these genes continued to increase in expression after day 42. As expected, genes that were downregulated (DOWN1, DOWN2, and DOWN3) during neuronal maturation were not significantly implicated in disease etiology or neuronal function.

Next, we performed transcriptomic analysis at three developmental time points (day 28, day 42, and day 56) using two

Fig. 2 *XPO7* LoF alters neuronal excitability by disrupting action potential regularities and dynamics. A Schematic workflow illustrating HD-MEA data processing, including spike sorting, extraction of activity and waveform features, and analysis of action potential propagation dynamics. **B** Principal component analysis of excitability features from control and *XPO7* LoF neurons across five independent differentiations. Genotype-specific clustering demonstrates strong reproducibility across batches, while clear separation between genotypes indicates robust discrimination of distinct excitability profiles. **C** Heatmap illustrating key electrophysiological measurements from HD-MEA assessments. Color scale indicates the relative change of each measurement, as annotated on the left. Outlined cells represent features significantly altered in $XPO7^{+/-}$ or $XPO7^{-/-}$ neurons compared to parental controls ($XPO7^{+/+}$). Asterisks denote levels of statistical significance. **D, E** Quantification of single-unit analysis results in $XPO7^{+/+}$ ($n = 23/5$), $XPO7^{+/-}$ ($n = 30/5$), and $XPO7^{-/-}$ neurons ($n = 31/5$), where n indicates the number of MEA cultures per genotype derived from 5 independent neuronal differentiation batches. **D** Activity metrics include spike amplitude and firing rate. **E** Waveform features include repolarization slope, recovery slope, and peak-to-valley duration; propagation metrics include maximum axonal length. **F** Heatmaps showing spike-triggered averaged signals across the electrode array for a representative neuron. The heatmap (left) displays spike amplitudes recorded at individual electrodes. Raw traces from the same neuron are shown on the right, with the three largest spike signals highlighted in red. **G** Sorted spikes from the same axonal tracking sequence of the representative neuron were extracted and temporally aligned. Spikes are arranged from top to bottom based on time of occurrence, with the three largest spike signals highlighted in red. **H** Correlation between spike latency and cumulative distance from the initiation site in the representative neuron shown in (**F**) and (**G**). The r^2 value was calculated using Pearson correlation. **I** Analysis of spike amplitude and cumulative distance during axonal conduction in $XPO7^{+/+}$ neurons at day 35. Spikes were grouped and binned based on their latency from the initiation site (bin width = 0.1 ms). For each tracked neuron, the mean spike amplitude was computed across all binned spikes, and the average across all neurons within a well was used to represent a single data point on the plot. **J** Quantification of spike amplitude in the distal axon at day 35. Spikes were grouped and binned based on their latency from the initiation site (bin width = 0.5 ms). The mean spike amplitude was computed for each bin per neuron, and the average across all neurons within a well was used to generate a single data point on the plot. (K) Heatmap showing changes in spike amplitude in $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons relative to controls in the distal axon from day 28 to day 56. Spikes were grouped and binned based on their latency from the initiation site (bin width = 0.1 ms). Color indicates the fold change in amplitude relative to the mean amplitude of age-matched control neurons. All statistical analyses were performed using ANOVA with post hoc Dunnett's multiple comparisons test. $*p < 0.05$; $**p < 0.01$; $***p < 0.001$; $****p < 0.0001$. Error bars represent SEM.

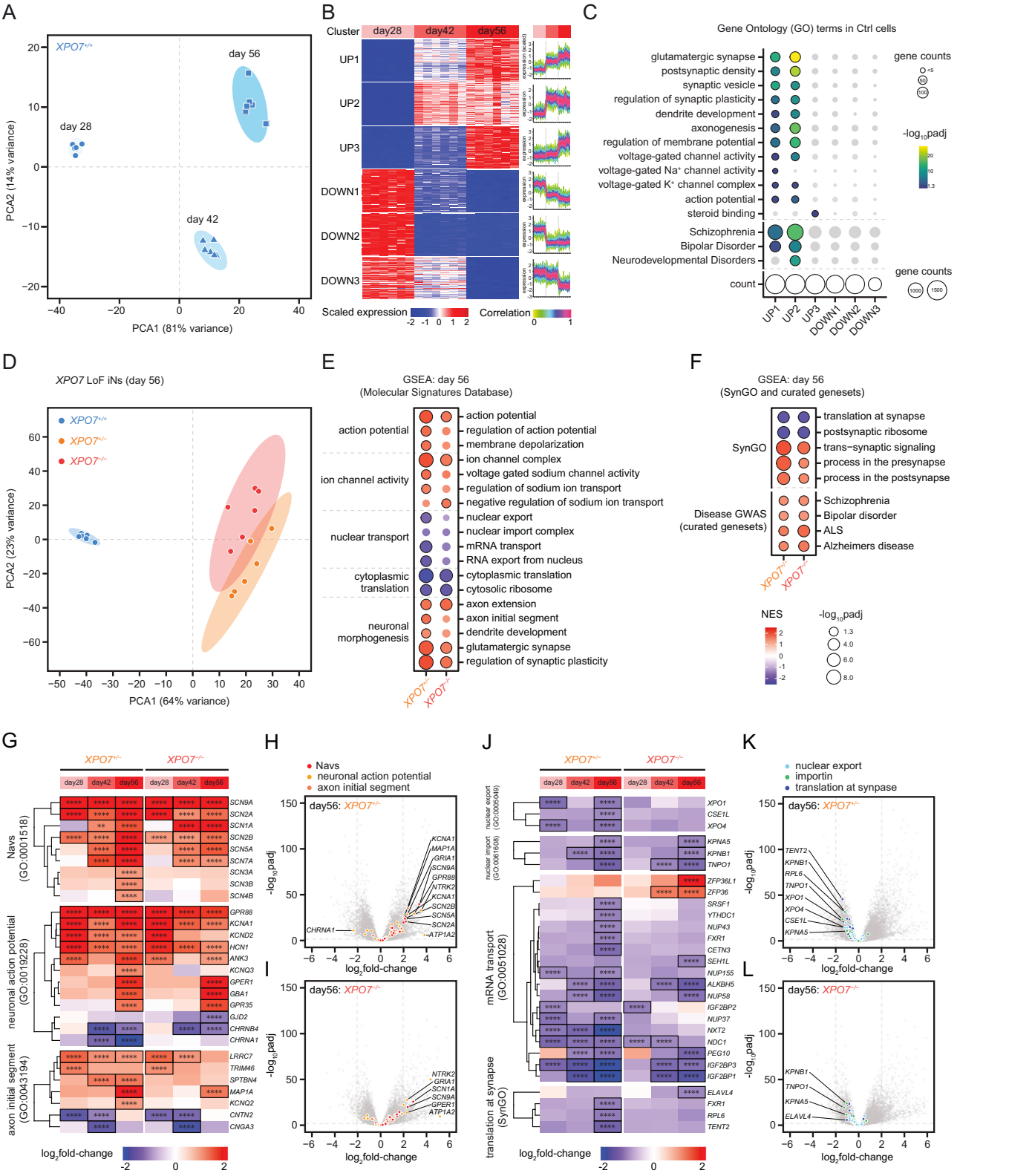
independent clones each of $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons, along with parental controls. All clones were differentiated in the same batch, and samples were collected at each time point for transcriptomic profiling. PCA at day 56, the time point at which significant alterations in ion channel activity and neuronal excitability were observed (Fig. 1 and 2), as well as at earlier time points, demonstrated clustering of *XPO7* LoF cell lines distinct from parental controls, suggesting substantial transcriptomic alterations caused by *XPO7* LoF (Fig. 3D and S5A). To investigate the affected molecular pathways, we performed gene set enrichment analysis (GSEA) using gene lists from MSigDB [37, 38], SynGO [39] and curated gene sets associated with neurological disorders identified by GWAS [40] (Fig. 3E-F and S5B-C). Consistent with our functional assays, transcriptomic analysis revealed significant alterations in the regulation of action potentials and ion channel activity in *XPO7* LoF neurons. We observed enrichment for upregulation of pathways associated with neuronal excitability and axon development, and enrichment for downregulation of pathways related to nuclear transport and cytoplasmic translation. Specifically, gene sets related to voltage-gated Na^+ channel activity (hereafter abbreviated as Na_v) were significantly enriched for upregulation in both $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons, with stronger enrichment observed in $XPO7^{+/-}$ neurons (Fig. 3E; $XPO7^{+/-}$: padj = 0.0149; $XPO7^{-/-}$: padj = 0.1015). Several genes encoding the alpha subunits of Na_v channels were significantly upregulated in both *XPO7* LoF cell types, including *SCN1A*, *SCN2A*, *SCN5A*, *SCN7A*, and *SCN9A* (Fig. 3G and 3H). In contrast, other genes, primarily encoding beta subunits, exhibited significant upregulation only in $XPO7^{+/-}$ cells, including *SCN3B*, *SCN4B*, and *SCN3A* (Fig. 3G and 3H), consistent with the more prominent increase in Na^+ channel density and action potential firing rate observed in $XPO7^{+/-}$ neurons (Fig. 1 and 2). In line with the increased axon length (Fig. 2E) and elevated action potential conduction velocity (Fig. S3F) detected by our single-cell axon tracking assay (Fig. 2C and K), gene sets associated with axonal morphogenesis, including axon extension and AIS development, were also significantly enriched for upregulation (Fig. 3G and H). Previous studies have identified *XPO7* as a nuclear exportin in various cell types, mediating the transport of specific cargo in a RanGTP-dependent manner [8]. In *XPO7* LoF neurons, gene sets related to nuclear transport, including both protein and mRNA transport, were significantly enriched for downregulation (Fig. 3E), and the top downregulated DEGs included other exportin family members such as *XPO1* and *XPO4* (Fig. 3J and K). Relatedly, gene sets associated with cytoplasmic translation were also consistently suppressed, implying broader cellular deficits that could be further exacerbated by impairments in nuclear transport and translation.

Using SynGO-annotated terms, we observed enrichment in gene sets related to synaptic function in *XPO7* LoF neurons, including enrichment for upregulation of trans-synaptic signaling pathways and enrichment for downregulation of synaptic translation (Fig. 3F and Fig. S5C). These findings suggest that *XPO7* LoF may differentially impact distinct synaptic processes at the transcriptomic level, warranting further investigation. In addition, we observed consistent and significant upregulation of gene sets linked to schizophrenia, bipolar disorder, amyotrophic lateral sclerosis (ALS), and Alzheimer's disease, identified by GWAS, broadly implicating a key role of *XPO7* in normal neuronal function.

The gene sets altered by *XPO7* LoF at day 56 were also examined across developmental stages (day 28 and day 42; Fig. 3G, J and S5E-F). We highlighted the top differentially expressed genes (DEGs) for each key gene set in heatmaps and presented all genes from selected gene sets at day 56 in volcano plots (Fig. 3H and K). All transcriptomic analyses were performed using data collected from two independent clones per genotype across all developmental stages (Fig. S5A-F). These data suggest major perturbations in neuronal function caused by *XPO7* LoF, characterized by upregulation of genes related to Na^+ channel activity and neuronal excitability, alongside downregulation of genes involved in nuclear export and cytoplasmic translation.

XPO7 LoF Disrupts Dendritic Architecture, Synaptic Density, and Subcellular Localization of Na_v Channels

We then investigated the functional impact of *XPO7* LoF on dendritic and synaptic morphology. Morphological analyses were performed at day 42 of differentiation (Fig. 4A), a time point at which our transcriptomic data began to reveal significant changes in pathways related to neuronal function (Fig. 3 and S5). To ensure robustness and reproducibility, we utilized a high-content imaging approach across three independent differentiations. For each differentiation, 25-36 imaging fields were acquired per well, and up to 20 wells were imaged per genotype. Further details on image processing and analysis are provided in the Methods



section. The mean value from all images of the same well represented one data point in the corresponding assay, allowing for consistent and unbiased statistical quantification.

Our results showed no differences across genotypes in the number of primary dendrites, which extend directly from the soma (Fig. 4B). However, a significant reduction in the number of dendritic segments branching from individual primary dendrites was observed in both *XPO7* LoF genotypes, along with a modest decrease in maximum dendritic length in *XPO7*$^{-/-}$ neurons. These findings were supported by the Sholl analysis, which indicated reduced dendritic complexity in *XPO7* LoF neurons. Together, these results suggest that *XPO7* LoF impairs dendritic architecture.

Fig. 3 *XPO7* LoF alters transcriptomic profiles across neuronal maturation and upregulates pathways related to Na^+ channel activity. **A** Principal component analysis of bulk RNA-seq data from control neurons at three different time points. **B** Heatmap showing the top 100 DEGs after hierarchical clustering of gene expression profiles in control neurons across three time points during neuronal maturation. Genes were clustered using fuzzy c-means clustering based on their expression trajectories across the three differentiation stages. **C** Gene Ontology (GO) enrichment analysis of RNA-seq data from control neurons. The size of each bubble indicates the number of gene counts. Bubbles filled with a gradient color represent pathways with significant enrichment (padj < 0.05), whereas bubbles filled with grey indicate pathways with nonsignificant enrichment (padj ≥ 0.05). **D** Principal component analysis of bulk RNA-seq data from *XPO7* LoF neurons at day 56. **E** Gene set enrichment analysis of bulk RNA-seq data from *XPO7* LoF neurons at day 56, using gene lists from the Molecular Signatures Database (MSigDB). The size of each bubble indicates the significance of pathway enrichment ($-\log_{10}$ converted). Bubbles outlined in black indicate significant pathways (padj < 0.05). Bubbles are filled with a gradient color to represent the GSEA normalized enrichment score (NES) of each pathway. **F** Gene set enrichment analysis of bulk RNA-seq data from *XPO7* LoF neurons at day 56, using gene lists from the Synaptic Gene Ontologies (SynGO) and curated gene sets associated with neurological disorders identified by GWAS. The size of each bubble indicates the significance of pathway enrichment ($-\log_{10}$ converted). Bubbles outlined in black indicate significant pathways (padj < 0.05). Bubbles are filled with a gradient color to represent the GSEA normalized enrichment score (NES) of each pathway. ALS, amyotrophic lateral sclerosis. **G** Heatmap showing the top DEGs for each key upregulated gene set across the three tested time points. Cells outlined in black indicate significant DEGs at the corresponding time point compared to parental controls. **H** and **I** Volcano plots showing all genes from selected upregulated gene sets at day 56. Top significant DEGs within the selected gene sets are annotated with gene symbols. The horizontal dashed line indicates padj $= 0.01$, and the vertical dashed lines indicate $\log_2$fold-change $= 2$ and -2. **J** Heatmap showing the top DEGs for each key downregulated gene set across the three tested time points. Cells outlined in black indicate significant DEGs at the corresponding time point compared to parental controls. **K** and **L** Volcano plots showing all genes from selected downregulated gene sets at day 56. Top significant DEGs within the selected gene sets are annotated with gene symbols. The horizontal dashed line indicates padj $= 0.01$, and the vertical dashed lines indicate $\log_2$fold-change $= 2$ and -2. See also Fig. S5.

Next, we quantified excitatory synapse density by performing immunostaining with presynaptic (Synapsin I/II) and postsynaptic (PSD-95) markers (Fig. 4C). To calculate synapse density for individual neurons, the total number of co-localized puncta along each dendritic segment was normalized to segment length, with dendritic morphology visualized using the MAP2 marker. We observed a significant reduction in synapse density in both *XPO7* LoF cell lines (Fig. 4D). These findings suggest that *XPO7* LoF leads to decreased synaptic density and reduced dendritic complexity, potentially resulting in reduced total connections between neurons in the network.

Given our electrophysiological findings of increased Na^+ channel activity and transcriptomic evidence of elevated expression of voltage-gated Na^+ channels (encoded by genes *SCN1A-SCN9A*; Fig. 3), we next examined the abundance and subcellular localization of Na^+ channel proteins. We performed immunostaining using a pan-Na_v antibody to detect voltage-gated Na^+ channels, alongside a specific antibody targeting $Na_v1.2$ (encoded by *SCN2A*; Fig. 4E). Consistent with previous reports, Na_v channels were expressed throughout the soma and enriched at the AIS. While *XPO7* LoF neurons exhibited normal Na_v channel enrichment at the AIS and normal AIS length (data not shown), they displayed increased levels of somatic Na_v channels, as labeled by either pan-Na_v or $Na_v1.2$ antibodies (Fig. 4F), consistent with the elevated Na^+ channel activity observed in the electrophysiological assays (Fig. 1 and 2).

While $Na_v1.2$ is well known for its role in action potential propagation and backpropagation, it has also been identified as a synapse-enriched protein by both SynGO and a recent large-scale synaptic proteomic study [39, 41]. In both rodent and human models, *SCN2A* LoF leads to reduced synaptic density and impaired network connectivity [42, 43]. We therefore examined synaptic $Na_v1.2$ expression in *XPO7* LoF neurons. In this assay, $Na_v1.2$ signal intensity within synaptic puncta colocalized with Synapsin I/II and PSD-95 was quantified as a measure of its synaptic localization (Fig. 4G). Notably, the synaptic $Na_v1.2$ level was significantly reduced in both *XPO7* LoF cell lines, in contrast to its elevated levels in the soma (Fig. 4H).

Taken together, these results indicate that *XPO7* LoF neurons exhibit reduced dendritic complexity and synapse density. Meanwhile, immunostaining revealed increased Na_v channel levels in the soma, accompanied by diminished levels at synapses. This mislocalization suggests impaired subcellular trafficking of Na_v channels.

XPO7 LoF Impairs Synaptic Proteome Composition and Na^+ channel Enrichment

Subsequently, we investigated the impact of *XPO7* LoF on the cytoplasmic and synaptic proteomic landscape of day 42 neurons using quantitative mass spectrometry. We first isolated the cytoplasmic fraction by removing nuclei using a commercially available protocol (Active Motif), followed by quantitative mass spectrometry to profile the cytoplasmic proteome. Consistent with the reduced synaptic density observed in *XPO7* LoF neurons (Fig. 4), GSEA of the cytoplasmic proteome revealed downregulation of GO terms related to synaptic function (e.g., postsynaptic density membrane, presynaptic active zone), as well as the mitochondrial envelope and nuclear export signal receptor activity (Fig. 5A). While Na^+ transmembrane transporter activity was selectively upregulated in $XPO7^{+/-}$ cells, overall ion transporter activity was reduced at the proteomic level, with particularly notable decreases in Ca^{2+} channel related terms, indicating impaired calcium signaling.

We next analyzed synaptic proteins and identified a greater number of downregulated differentially expressed proteins (DEPs, defined as padj < 0.05 and $\log_2$fold-change < -0.1 or > 0.1) in both $XPO7^{+/-}$ neurons (227 downregulated vs. 108 upregulated) and $XPO7^{-/-}$ neurons (237 downregulated vs. 154 upregulated; Fig. 5B-D). Among these DEPs, several canonical synaptic markers, including PSD-95, Synaptotagmin-1, VGLUT1, and VGLUT2, were consistently downregulated (Fig. 5D-G and S6A-D), corroborating the reduced synapse density (Fig. 4) and further supporting a role for *XPO7* in maintaining neuronal connectivity. The majority of these synaptic markers were downregulated in both $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons (Fig. 5D), indicating that haploinsufficiency of *XPO7* is sufficient to impair synaptic integrity in excitatory neurons. Notably, NPTX2 exhibited consistent and pronounced downregulation in both *XPO7* LoF lines ($XPO7^{+/-}$: $\log_2$fold-change $= -0.76 \pm 0.20$; $XPO7^{-/-}$: $\log_2$fold-change $= -1.36 \pm 0.05$), with a magnitude exceeding that of canonical synaptic markers (Fig. 5G, S6B, and D). NPTX2 has previously been reported to be reduced in cerebrospinal fluid and prefrontal cortex synaptic fractions from individuals with schizophrenia [44, 45], suggesting that *XPO7* LoF-mediated downregulation of NPTX2 may contribute to convergent molecular mechanisms underlying disease-associated synaptic dysfunction. Overall, cytoplasmic proteome analysis revealed broad downregulation of pathways and proteins related to synaptic function, reflecting at least in part the reduced synapse density in *XPO7* neurons.

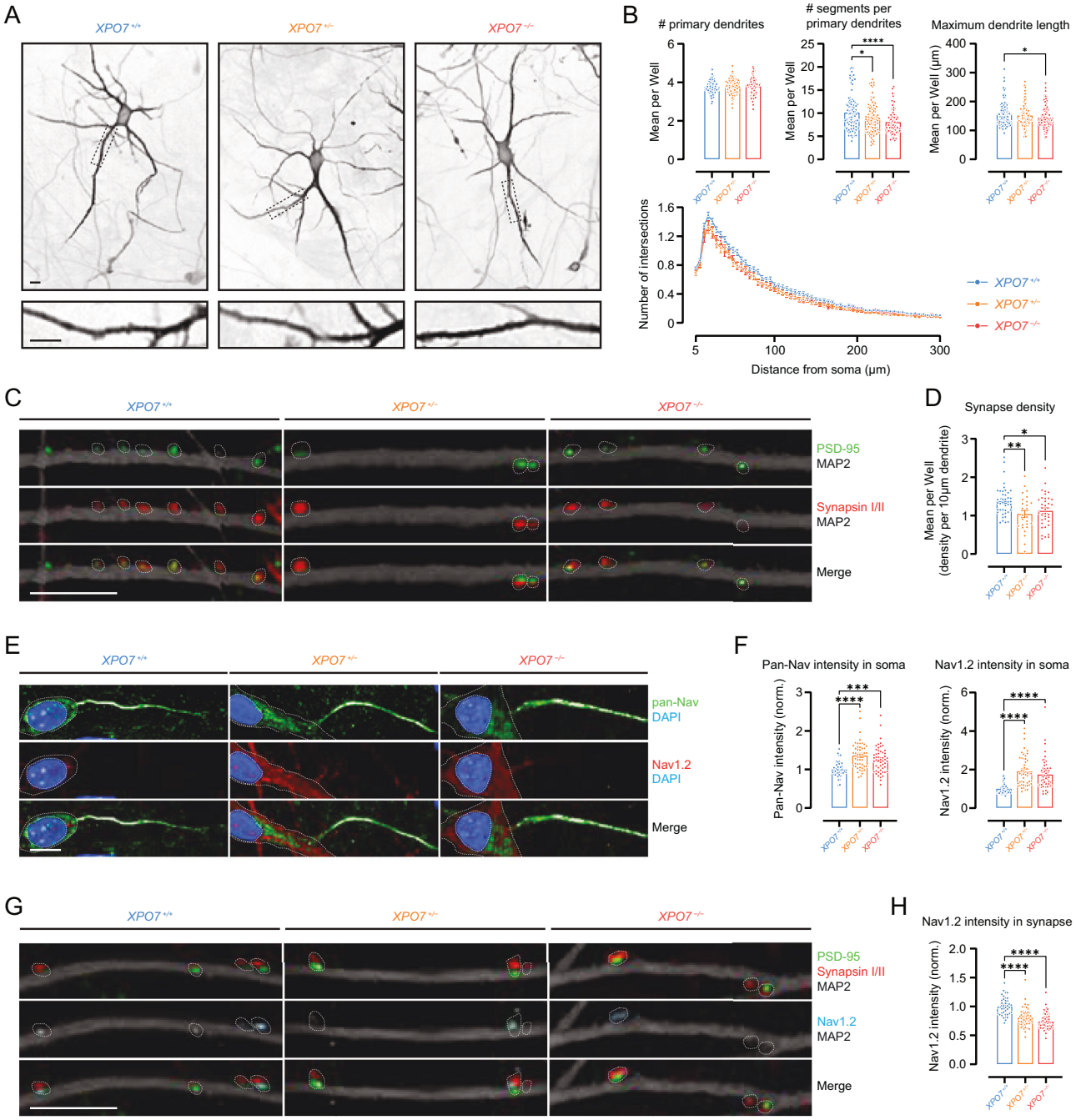
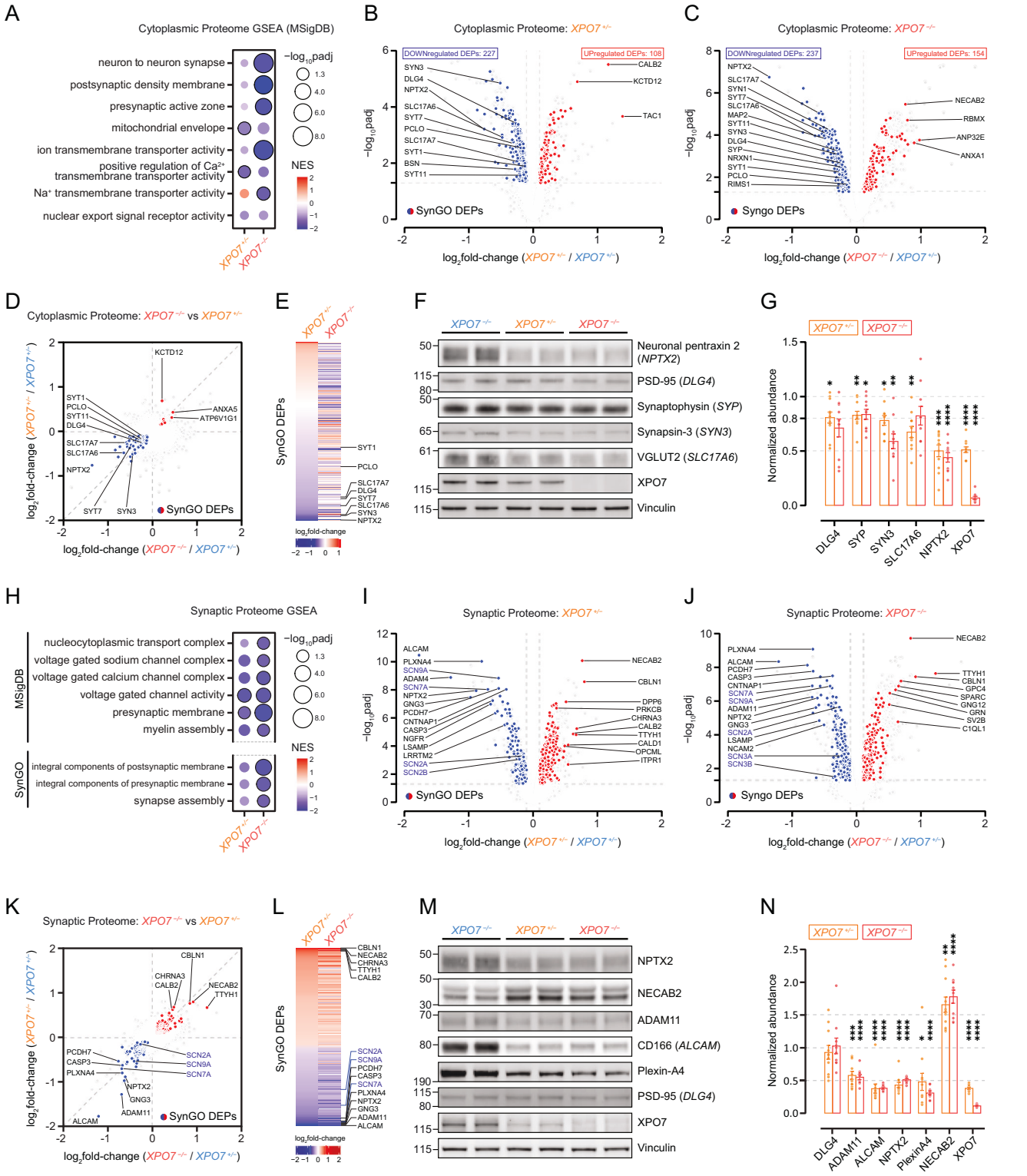


Fig. 4 *XPO7* LoF reduces dendritic complexity, decreases synaptic density, and alters Na_v channel subcellular localization. **A** Representative immunofluorescent images of day 42 *XPO7*$^{+/+}$, *XPO7*$^{+/-}$, *XPO7*$^{-/-}$ neurons stained with the dendritic marker MAP2. Black dashed boxes in the upper images indicate the regions that are cropped and magnified in the bottom images. Scale bar, 10 µm. **B** Morphological analysis of *XPO7*$^{+/+}$ ($n = 101$), *XPO7*$^{+/-}$ ($n = 99$), and *XPO7*$^{-/-}$ ($n = 91$) neuronal cultures by quantifying the number of primary dendrites, the number of dendritic segments per primary dendrite, and the maximum dendrite length. Distribution plot showing Sholl intersections relative to distance from the soma. **C** Representative immunofluorescent images of day 42 *XPO7*$^{+/+}$, *XPO7*$^{+/-}$, and *XPO7*$^{-/-}$ neurons stained with presynaptic (Synapsin I/II), postsynaptic (PSD-95), and dendritic (MAP2) markers. Scale bar, 10 µm. **D** Quantification of synapse density in *XPO7*$^{+/+}$ ($n = 41$), *XPO7*$^{+/-}$ ($n = 26$), and *XPO7*$^{-/-}$ ($n = 36$) neuronal cultures by calculating the total number of colocalized puncta along each dendritic segment, normalized to segment length. **E** Representative immunofluorescent images of day 42 *XPO7*$^{+/+}$, *XPO7*$^{+/-}$, and *XPO7*$^{-/-}$ neurons stained with a pan-Na_v antibody to detect voltage-gated Na^+ channels, along with a specific antibody targeting $Na_v1.2$. Scale bar, 10 µm. **F** Quantification of the relative fluorescent intensity of pan-Na_v and $Na_v1.2$, with the intensity of *XPO7*$^{+/-}$ ($n = 48$) and *XPO7*$^{-/-}$ ($n = 52$) neuronal cultures normalized to that of *XPO7*$^{+/+}$ ($n = 39$) neuronal cultures, respectively. **G** Representative immunofluorescent images of day 42 *XPO7*$^{+/+}$, *XPO7*$^{+/-}$, and *XPO7*$^{-/-}$ neurons stained with antibodies targeting $Na_v1.2$ and synaptic markers (Synapsin I/II and PSD-95). Scale bar, 10 µm. **H** Quantification of the relative fluorescent intensity of $Na_v1.2$ within puncta colocalized with Synapsin I/II and PSD-95, with the intensity of *XPO7*$^{+/-}$ ($n = 54$) and *XPO7*$^{-/-}$ ($n = 51$) neuronal cultures normalized to that of *XPO7*$^{+/+}$ ($n = 39$) neuronal cultures, respectively. All statistical analyses were performed using ANOVA with post hoc Dunnett's multiple comparisons test. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Error bars represent SEM.



In order to clarify the impact of *XPO7* LoF on existing synapses, we isolated the crude synaptoneurosome fraction and profiled the synaptome using quantitative mass spectrometry. We adapted a biochemical approach for isolating crude synaptoneurosome fractions from iPSC-derived neurons, based on published protocols with minor modifications [46]. GSEA using MSigDB as a reference indicated consistent downregulation of pathways related to the nucleocytoplasmic transport complex (GO:0031074) and myelin assembly (GO:0032288; Fig. 5H). In addition, GSEA using SynGO gene sets further revealed consistent downregulation of pathways related to integral components of both presynaptic (GO:0099056) and postsynaptic (GO:0099055) membranes. Notably, pathways associated with the voltage-gated Na^+ channel complex (including $Na_V1.2$ (*SCN2A*), $Na_V1.7$ (*SCN9A*), $Na_V1.3$ (*SCN3A*), SCN2B (*SCN2B*), and SCN3B (*SCN3B*)) were consistently downregulated in both *XPO7* LoF cell lines (Fig. 5I-

Fig. 5 *XPO7* LoF impairs synaptic composition and reduces Na^+ channel abundance in the synaptic proteome. A GSEA of cytoplasmic proteomic data from *XPO7* LoF neurons at day 42, using gene lists from MSigDB. The size of each bubble indicates the significance of pathway enrichment (-$\log_{10}$ of padj). Bubbles outlined in black indicate significantly enriched pathways (padj < 0.05). Bubbles are filled with a gradient color representing the GSEA normalized enrichment score of each pathway. **B** and **C** Volcano plots showing all synaptic proteins annotated by SynGO. Top significant DEPs are labeled with the gene symbol of the corresponding protein. The horizontal dashed line indicates padj = 0.05, and the vertical dashed lines indicate $\log_2$fold-change thresholds of 0.1 and -0.1. **D** Correlation of protein $\log_2$fold-changes between $XPO7^{+/-}$ and $XPO7^{-/-}$ cytoplasmic proteomes. Top significant DEPs are labeled with the gene symbol of the corresponding protein. **E** Heatmap showing $\log_2$fold-changes in the cytoplasmic proteomes of $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons. **F** Cytoplasmic extracts were analyzed by Western blot to assess top DEPs identified by cytoplasmic proteomics, including NPTX2, PSD-95, Synaptophysin, Synapsin-3, VGLUT2, and XPO7, with Vinculin used as a loading control. **G** Quantification of (**F**) by normalizing the intensity of probed targets in $XPO7^{+/-}$ ($n = 11$) and $XPO7^{-/-}$ ($n = 10$) neurons to the mean intensity of the corresponding $XPO7^{+/+}$ ($n = 6$) control group. Complete data can be found in Fig. S6B and D. **H** GSEA of synaptic proteomic data from *XPO7* LoF neurons at day 42, using gene lists from MSigDB and SynGO. The size of each bubble indicates the significance of pathway enrichment (-$\log_{10}$ of padj). Bubbles outlined in black indicate significantly enriched pathways (padj < 0.05). Bubbles are filled with a gradient color representing the GSEA normalized enrichment score of each pathway. **I** and **J** Volcano plots showing all synaptic proteins annotated by SynGO. Top significant DEPs are labeled with the gene symbol of the corresponding protein. The horizontal dashed line indicates padj = 0.05, and the vertical dashed lines indicate $\log_2$fold-change thresholds of 0.1 and -0.1. **K** Correlation of protein $\log_2$fold-changes between $XPO7^{+/-}$ and $XPO7^{-/-}$ synaptic proteomes. Top significant DEPs are labeled with the gene symbol of the corresponding protein. **L** Heatmap showing $\log_2$fold-changes in the synaptic proteomes of $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons. **M** Synaptic fractions were analyzed by Western blot to assess top DEPs identified by synaptic proteomics, including NPTX2, NECAB2, ADAM11, CD166, Plexin-A4, and XPO7. PSD-95 and Vinculin served as loading controls. **N** Quantification of (**M**) by normalizing the intensity of probed targets in $XPO7^{+/-}$ ($n = 11$) and $XPO7^{-/-}$ ($n = 10$) neurons to the mean intensity of the corresponding $XPO7^{+/+}$ ($n = 6$) control group. Complete data can be found in Fig. S6F and H. All statistical analyses were performed using ANOVA with post hoc Dunnett's multiple comparisons test. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Error bars represent SEM.

L). This is consistent with the reduced synaptic protein level of $Na_v1.2$ (Fig. 4), indicating a relative depletion of Na^+ channels at synapses despite elevated gene expression and increased protein abundance in the soma. Additionally, we observed a drastic reduction of NPTX2 in the synaptic fractions ($XPO7^{+/-}$: $\log_2$ fold-change = -0.89 ± 0.04; $XPO7^{-/-}$: $\log_2$ fold-change = -1.61 ± 0.05), indicating reduced NPTX2 levels at the existing synapses (Fig. 5M-N). The top DEPs from our proteomic analysis were independently validated using biochemical assays (Fig. 5F, M and S6A-H), as demonstrated by correlation comparisons showing high consistency, supporting the accuracy and reliability of our proteomic analysis (Fig. S6I). All proteomic and biochemical validation experiments were performed using two independent clones of $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons (Fig. S6A-H).

Taken together, cytoplasmic proteomic analysis suggests an overall reduction in synaptic proteins in *XPO7* LoF neurons, as evidenced by a general decrease in canonical synaptic markers. Further analysis of the synaptome revealed alterations in existing synapses, including dysregulation of pathways related to Na_v channel conductance and synapse assembly. These findings indicate a deficiency of local Na^+ channels at synapses and suggest that electrochemical signaling is impaired in *XPO7* LoF synapses.

XPO7 LoF Disrupts Neuronal Network Synchrony

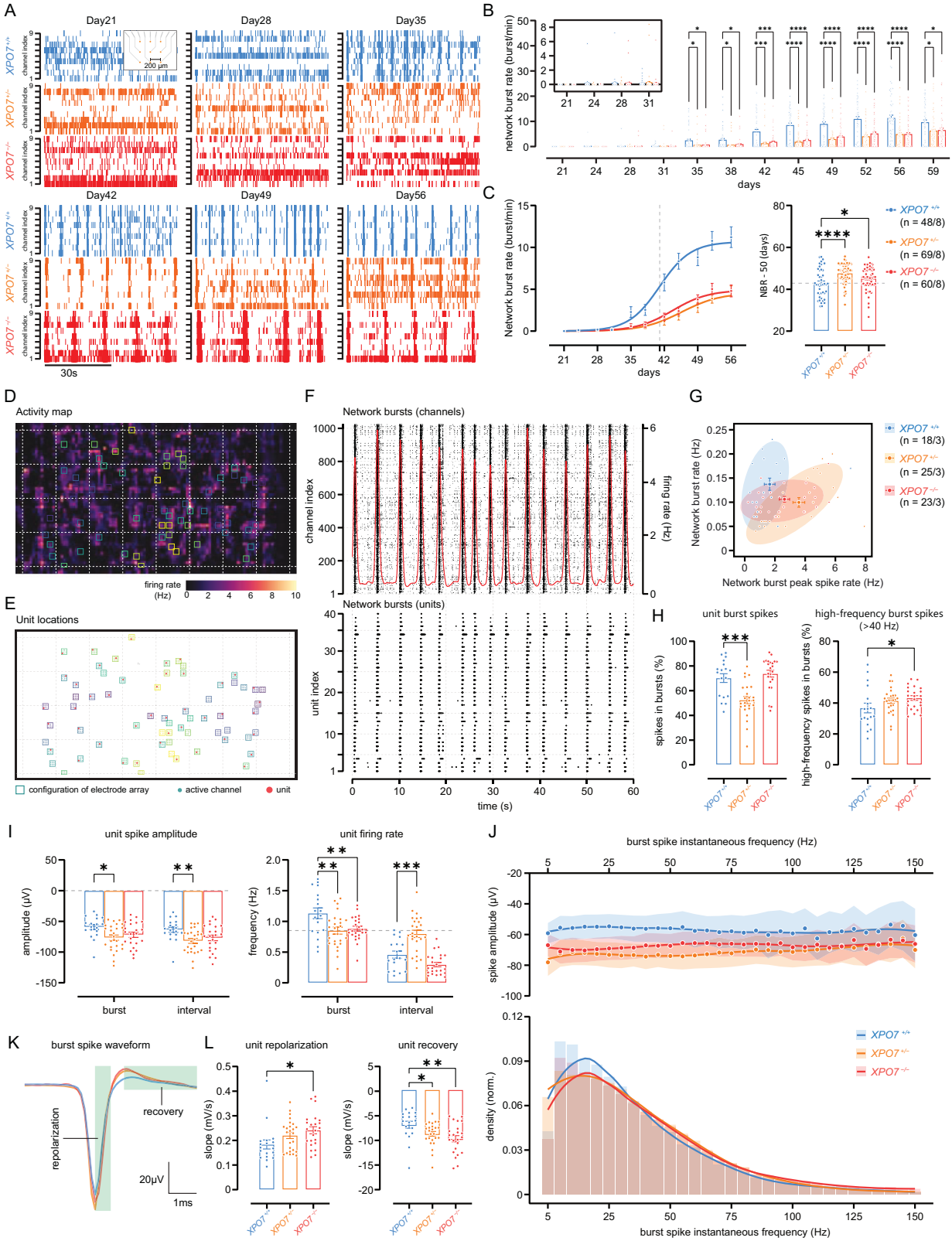
Next, we examined the impact of *XPO7* LoF on overall network connectivity and synchrony. Using a nine-electrode (low-density) microelectrode array system with fixed 200 μm electrode spacing (Multi Channel Systems), we first monitored network activity across different stages of neural differentiation (Fig. 6A and S7A). During the first four weeks, neuronal activity was primarily characterized by isolated, asynchronous spikes and bursts (high-frequency trains of action potentials). As the network integrated, neuronal activity gradually transitioned into rhythmic and synchronized bursts across the neural network (Fig. 6A). We repeated these experiments using 48-69 neuronal cultures/networks per genotype derived from eight independent differentiations, enabling the detection of stable and reproducible genotype-dependent differences.

To quantify network dynamics, we calculated the network burst rate (NBR), defined as the number of network bursts per minute. Neuronal networks derived from parental control lines reached half-maximal NBR (NBR-50) by day 42 of differentiation (Fig. 6B-C). Compared to controls, *XPO7* LoF networks exhibited significantly and consistently reduced NBRs across developmental stages (Fig. 6B). Moreover, both *XPO7* LoF lines showed a modest but significant delay in achieving NBR-50 ($XPO7^{+/-}$: ΔNBR-50 = +4.79; $XPO7^{-/-}$: ΔNBR-50 = +2.30; Fig. 6C), suggesting impaired network integration. Consistent with reduced NBRs, *XPO7* LoF neurons exhibited altered network burst duration (Fig. S7B) and an increased number of spikes per network burst at later stages (Fig. S7C), indicating broader disruptions in network burst structure and maturation. Together, these results suggest that *XPO7* LoF impairs the development and temporal organization of synchronous network activity.

To assess neuronal connectivity with higher spatial and temporal precision, we used HD-MEA recordings (Fig. 2), which resolved the network synchronization at both population and single-unit levels. We repeated these experiments using 18-25 neuronal cultures/networks per genotype derived from three independent differentiations. At day 42 post-differentiation, corresponding to the NBR-50 stage, HD-MEA recordings captured robust, synchronized network activity across all genotypes (Fig. 6D-F), confirming findings from low-density MEAs with enhanced precision. Notably, this analysis revealed a significant reduction in NBRs in *XPO7* LoF neurons (73% and 77% of control neurons for $XPO7^{+/-}$ and $XPO7^{-/-}$, respectively), accompanied by elevated peak spike rates (Fig. 6G).

To resolve activity at the single-unit level, spikes were sorted, and individual units were identified (Fig. 6E). The activity of sorted units was temporally aligned with network bursts detected in the unsorted population signal (Fig. 6F). In control networks, 70% of spikes occurred within network bursts, compared to only 52% in $XPO7^{+/-}$ neurons ($p = 0.0003$; Fig. 6H). We then examined whether the temporal structure of spike generation within bursts was altered. A greater proportion of network burst spikes occurred at high instantaneous frequencies exceeding 40 Hz ($XPO7^{+/+}$: 37%; $XPO7^{+/-}$: 41%; $XPO7^{-/-}$: 43%; Fig. 6H). This shift suggests altered temporal dynamics of spike generation during network synchronization, potentially driven by dysregulated ion channel function.

We next examined activity features of sorted units by extracting spikes occurring within network burst and interburst interval (IBI) windows, referred to as "burst" and "interval" in Fig. 6, across genotypes. This analysis was performed at day 42, as early-stage networks (before NBR-50) exhibited only sparse bursts, which prevented reliable extraction of burst-specific kinetic features. *XPO7* LoF neurons displayed larger spike amplitudes both during and outside of bursts (Fig. 6I, left panel and S7D), in agreement



with findings from Fig. 2. Conversely, spike firing rates during network bursts were reduced in both *XPO7* LoF lines (Fig. 6I, right panel). Interestingly, *XPO7*$^{+/-}$ neurons exhibited significantly elevated firing rates during interburst intervals (Fig. 6I), reaching levels comparable to those observed during bursts, further supporting disrupted temporal regulation of the network. These physiological deficits mirror the molecular synaptome alterations observed in *XPO7* LoF neurons (Fig. 4 and Fig. 5). To further characterize firing dynamics, burst spikes were binned by instantaneous firing frequency and analyzed for density

Fig. 6 *XPO7* LoF disrupts the temporal precision and regularity of neuronal network activity by altering firing rate dynamics and spike waveform properties. A Representative 1-minute raster plots of electrophysiological activity from $XPO7^{+/+}$, $XPO7^{+/-}$, and $XPO7^{-/-}$ neuronal networks at multiple developmental time points. A multielectrode array system with 9 electrodes per array and fixed 200 μm electrode spacing was used to monitor network activity across different stages of neural differentiation and maturation. **B** Network synchrony across developmental time. Each point represents the mean network burst rate (number of network bursts per minute) for a single neuronal network (one MEA chip) at the indicated time point in $XPO7^{+/+}$ ($n = 48/8$), $XPO7^{+/-}$ ($n = 69/8$), and $XPO7^{-/-}$ ($n = 60/8$) neuronal cultures, where n indicates the number of MEA networks per genotype derived from 8 independent neuronal differentiation batches. Neuronal activity for each included network was recorded over a 59-day period, with two recordings performed per week. **C** Summary of the half-maximal network burst rate (NBR-50) values derived from sigmoid curve fitting of network burst rate trajectories. **D** Representative activity map of neuronal networks at day 42. Heatmap shows the mean firing rates recorded from 6,600 electrodes spaced at 35 μm pitch over a 120-second recording period. **E** Locations of sorted units from the example neuronal network shown in (**D**). **F** Representative 1-minute raster plots of electrophysiological activity from the example neuronal network shown in (**D**). The top plot shows spike time stamps from 1000 simultaneously recorded channels, with red traces indicating the accumulated firing rates across all channels. The bottom plot shows spike time stamps from sorted single units. All plots were sampled from the same 1-minute recording and share the same x-axis and time scale. **G** Network synchrony assessed by measuring network burst rate and peak spike rate using HD-MEA. Each point represents synchrony features from a single neuronal network in $XPO7^{+/+}$ ($n = 18/3$), $XPO7^{+/-}$ ($n = 25/3$), and $XPO7^{-/-}$ ($n = 23/3$) neuronal cultures, where n indicates the number of MEA networks per genotype derived from 3 independent neuronal differentiation batches. **H** Summary of spike distributions from sorted neurons. The left graph shows the percentage of spikes that occurred within network burst windows. The right graph shows the percentage of burst spikes with instantaneous firing frequencies exceeding 40 Hz. **I** Summary of spike amplitude and firing rate of sorted neurons. Spikes from each sorted neuron were grouped based on whether they occurred during network bursts ("burst") or interburst intervals ("interval"). Each point represents the mean amplitude (left) or mean firing rate (right) of all sorted neurons within the same neuronal network. **J** Summary of burst spike regularity based on spike distribution and amplitude across frequency bins. Burst spikes were grouped by their instantaneous firing frequencies, and the density and mean amplitude of each bin were calculated for every sorted neuron. Temporal distributions of binned spikes were compared at the group level using the Anderson-Darling k-sample test. Both $XPO7^{+/-}$ and $XPO7^{-/-}$ groups exhibited statistically significant differences in spike density distributions compared to the $XPO7^{+/+}$ group (AD test statistics = -0.84 and -0.39, respectively; significance threshold $\alpha = 0.25$). **K** Representative burst spike waveforms from $XPO7^{+/+}$, $XPO7^{+/-}$, and $XPO7^{-/-}$ neurons. **L** Summary of waveform feature analysis of burst spikes. Repolarization slope (left) and recovery slope (right) were extracted from the averaged burst spike waveforms of sorted neurons. All statistical analyses were performed using ANOVA with post hoc Dunnett's multiple comparisons test. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Error bars represent SEM. See also Fig. S7.

distributions (Fig. 6J). This revealed a marked shift in *XPO7* LoF neurons, with reduced normalized density of low-frequency spikes (< 40 Hz) and increased density of high-frequency spikes (≥40 Hz). Despite these shifts, the mean spike amplitude within each frequency bin remained consistently different, with *XPO7* LoF neurons exhibiting elevated amplitudes across all frequencies. These analyses provided high-resolution insights into the temporal organization of neuronal activity, demonstrating that *XPO7* LoF disrupts the balance between synchronous and asynchronous firing by reducing the temporal precision of burst-associated spiking.

To further investigate the cellular substrate of these irregularities, we extracted spike waveforms occurring within or between network bursts and analyzed the averaged signal characteristics for each sorted unit (Fig. 6K). This analysis revealed a marked increase in repolarization slope (Fig. 6L and S7H), a decrease in recovery slope (Fig. 6L and S7I), and shorter peak-to-valley and half-width durations of burst spikes (Fig. S7E-G) in *XPO7* LoF neurons, all indicative of accelerated action potential kinetics. These waveform features align with the elevated instantaneous firing frequencies observed in *XPO7* LoF neurons during network bursts.

Together, these findings suggest that *XPO7* LoF neurons show dysregulation of network synchronization, including changes in burst spike frequency and waveform properties. These alterations disrupt network burst structure and regularity, ultimately impairing the temporal precision and coordination of network synchronization.

DISCUSSION

Loss of function of *XPO7* has been implicated in increased risk for schizophrenia [7, 15]. Utilizing an iPSC-derived neuron model, we demonstrate that both heterozygous and homozygous *XPO7* LoF lead to increased Na^+ channel availability and altered voltage-dependent activation (Fig. 1), resulting in dysregulated neuronal excitability (Fig. 2). Transcriptomic (Fig. 3) and proteomic (Fig. 5) analyses further reveal widespread disruptions in pathways regulating ion channel function, synaptic integrity, and nuclear export. In particular, impaired Na^+ channel dynamics (Fig. 1) and aberrant subcellular distribution at the synapse (Fig. 4) may underlie the disruption of network synchrony and connectivity (Fig. 6). Together, these findings highlight the critical role of *XPO7* in maintaining neuronal homeostasis and suggest that electrophysiological and structural abnormalities arising from *XPO7* LoF in glutamatergic neurons may contribute to schizophrenia pathophysiology.

While we have focused on the consistent impairments in both $XPO7^{+/-}$ and $XPO7^{-/-}$ neurons, not all phenotypes were uniformly observed across both genotypes. Increased Na^+ channel current density and elevated firing rates were more prominent in $XPO7^{+/-}$ neurons, whereas consistent and dosage-dependent effects were evident for Na^+ channel inactivation, accelerated action potential conduction kinetics, reduced network synchronization, and broader changes in the synaptome, with more pronounced alterations in $XPO7^{-/-}$ neurons. These differences may reflect gene dosage sensitivity and/or compensatory mechanisms engaged in response to complete *XPO7* loss. However, our transcriptomic data did not reveal significant upregulation of other *XPO* exportins, which instead showed a consistent trend of downregulation in both *XPO7* LoF genotypes (Fig. S8). In contrast, proteomic analyses of cytoplasmic, nuclear, and synaptic fractions consistently demonstrated significant upregulation of Calreticulin (encoded by *CALR*) specifically in $XPO7^{-/-}$ neurons. Calreticulin is a multifunctional protein involved in calcium homeostasis, protein folding, and nuclear export. Its role in nuclear export involves binding cargo proteins such as NR3C1 through a mechanism similar to that of the XPO family, requiring a nuclear export signal (NES) and RanGTP [47]. Rare functional mutations in the *CALR* promoter region, such as the -220A variant that increases gene expression, have been identified in individuals with psychiatric disorders [48]. Conversely, the -205T mutation, which decreases Calreticulin expression, has been reported in a case of schizoaffective disorder, suggesting that both upregulation and downregulation of Calreticulin expression may contribute to psychiatric genetics [49]. These results suggest that Calreticulin upregulation

in *XPO7*$^{-/-}$ neurons may represent a compensatory adaptation to disrupted nuclear export in the context of complete *XPO7* loss and highlight its potential role as a modifier of disease-relevant phenotypes. Similar genotype-dependent phenotypic differences have been recognized in models of neurodevelopmental and neuropsychiatric genetics, where homozygous null mutations can trigger developmental compensation that attenuates or alters the phenotype relative to the heterozygous state. A pertinent example is *SHANK3* knockout mice, which, despite exhibiting marked neurodevelopmental abnormalities, display compensatory upregulation of other synaptic proteins, including the homologues Shank1 and Shank2, in the brain [50]. A similar pattern is observed in *CACNA1A* knockout mice, where complete loss of this gene induces compensatory upregulation of other Ca^{2+} channel subtypes, partially ameliorating cerebellar dysfunction and yielding a qualitatively distinct rather than an exaggerated heterozygous phenotype [51]. These lines of evidence suggest that the effects of a heterozygous mutation do not necessarily predict those of the homozygous condition, as nonlinear compensatory mechanisms may arise.

We leveraged two electrophysiological platforms, SyncroPatch 384 and HD-MEA, to uncover the impact of *XPO7* LoF on both cell-autonomous and network-level neuronal function. The SyncroPatch 384 system enabled systematic measurement of ionic conductance across large populations of iPSC-derived neurons, providing the sample size necessary for robust analyses and significantly accelerating data acquisition compared to traditional manual patch-clamp methods [29]. In prior studies, high-throughput automated patch-clamp (APC) technology has been predominantly applied to iPSC-derived non-neuronal models, such as cardiomyocytes [52], owing to the substantial challenges associated with achieving consistent neuronal differentiation, promoting functional maturation, and minimizing electrophysiological variability. While recent studies have extended APC applications to iPSC-derived neuronal models, including sensory and dopaminergic neurons [53, 54], our work represents the first application of APC to disease-relevant neuronal models for characterizing the electrophysiological effects of schizophrenia-associated risk genes in functionally differentiated excitatory neurons. By using the SyncroPatch 384 system, we identified significant alterations in Na^+ channel properties in *XPO7* LoF neurons (Fig. 1), including increased current densities, a hyperpolarizing shift in voltage-dependent activation, reduced frequency-dependent inactivation, and an increased number of Na^+ peaks in response to voltage ramps. In contrast to SyncroPatch, the HD-MEA platform captures electrical activity from up to 26,400 microelectrodes, with 1,024 channels available for simultaneous readout of extracellular voltages across the neuronal culture. By integrating spike sorting and single-unit feature extraction, this approach enables analysis of both single-cell activity and population-level dynamics within complex neuronal networks [31, 32]. This dual capability provides high spatiotemporal resolution for characterizing neuronal excitability and network behavior across multiple scales. This technology has previously been utilized in human models to investigate the pathophysiology of epilepsy, ALS, and FTLD [33, 55]. Our HD-MEA recordings revealed that *XPO7* LoF induces persistent and developmentally progressive changes in neuronal excitability and network synchrony. Key excitability features, including spike amplitude, firing rate, and waveform dynamics, diverged from controls as early as day 28 and continued to intensify through day 56 (Fig. 2). Similarly, network-level properties such as burst rate and temporal coordination showed early impairments that became more pronounced during maturation (Fig. 6). These dynamic electrophysiological changes were paralleled by transcriptomic shifts observed across the same developmental time points (Fig. 3). Specifically, genes related to Na^+ channel activity and axonal development were upregulated over time in *XPO7* LoF neurons, consistent with increased excitability and extended axonal arborization observed at the functional level. Our data revealed disruptions in both activity and regularity at the network, cellular, and subcellular levels, providing a mechanistic link from altered Na^+ channel availability to changes in single-unit action potential waveforms, neuronal excitability, and ultimately, network synchronization. These findings highlight the power of precision electrophysiological platforms to capture both cell-autonomous and network-level alterations with high temporal and spatial precision.

Synaptic deficits have long been implicated in the pathophysiology of schizophrenia and other neuropsychiatric disorders [56, 57]. Genetic studies have implicated synaptic pathways in schizophrenia, identifying numerous risk loci enriched for genes involved in synaptic organization and transmission [5, 6]. By integrating morphological, biochemical, and functional assays, we identified significantly reduced synapse density, altered synaptic proteomic composition, and impaired network synchrony in *XPO7* LoF neurons (Fig. 4–6). Postmortem studies consistently reveal decreased levels of the presynaptic and postsynaptic markers, including reduced synaptophysin in the frontal cortex, cingulate cortex, and hippocampus, as well as diminished dendritic spine density and postsynaptic density proteins in cortical regions [56]. Although these findings may be influenced by methodological variations and confounding factors, such as antipsychotic exposure and postmortem intervals, the overall evidence supports synaptic alterations in schizophrenia. These synaptic impairments are recapitulated in novel models such as human neurons derived from schizophrenia patients or schizophrenia-discordant twins, which exhibit increased neuronal excitability, reduced morphological complexity, dysregulation of synapse-related genes, decreased synaptic density, and functional abnormalities including impaired neuronal connectivity [16, 17, 19]. *XPO7* was implicated in schizophrenia through a recent exome sequencing meta-analysis (SCHEMA), which identified ten high-confidence genes associated with the disorder at exome-wide significance [7]. Our findings demonstrate that *XPO7* LoF leads to significant structural, molecular, and functional synaptic deficits, aligning with human genetic evidence and highlighting a mechanistic link between *XPO7* dysfunction and synaptic pathology in schizophrenia. Future studies should focus on elucidating how the altered molecular composition of the synaptome under *XPO7* LoF contributes to aberrant synaptic function and deficient network synchronization. One potential target is NPTX2, which has previously been reported to be reduced in the cerebrospinal fluid and prefrontal cortex synaptic fractions of individuals with schizophrenia. Its significant downregulation observed in our study (*XPO7*$^{+/-}$ at day 42: transcriptomic $\log_2$fold-change = -0.32 ± 0.08; cytoplasmic proteomic $\log_2$fold-change = -0.75 ± 0.12; synaptic proteomic $\log_2$fold-change = -0.85 ± 0.07) suggests a potential convergence of molecular pathways disrupted by both *XPO7* and *NPTX2* loss of function.

Our findings further point to altered Na^+ channel dynamics as a potential pathophysiological feature that converges across diverse models of schizophrenia. A recent study of neurons derived from individuals with high polygenic risk scores for schizophrenia reported increased Na^+ channel availability and altered channel kinetics, mirroring our findings in *XPO7* LoF neurons and suggesting that Na^+ channel dysregulation may contribute to schizophrenia risk associated with common variants [20]. Notably, in the SCHEMA study, where *XPO7* was first implicated as a schizophrenia risk gene, *SCN2A*, encoding the $Na_v1.2$ channel, was also identified with a substantial odds ratio of 5.35 [7]. Together, these findings suggest that both common and rare genetic variants may converge on impaired Na^+ channel function as a shared mechanism in the pathophysiology of schizophrenia. The elevated levels of Na^+ channels in *XPO7* LoF neurons were further validated by transcriptomic analysis, which revealed a progressive

upregulation of a cluster of genes encoding Na_v channels ($Na_v1.1$, $Na_v1.2$, $Na_v1.3$, $Na_v1.5$, and $Na_v1.7$; Fig. 3), indicating a sustained increase in Na^+ channel expression during maturation. This finding was further corroborated by immunofluorescence analyses, which demonstrated increased somatic expression and clustering of Na_v1 channels (Fig. 4). Most strikingly, alongside the elevated Na^+ channel availability observed in the soma and increased neuronal excitability (Fig. 1–4), we detected reduced levels of $Na_v1.2$ at the synapse, accompanied by impaired network synchrony (Fig. 4–6). These results suggest that the altered expression and subcellular distribution of Na^+ channels caused by *XPO7* LoF are associated with the deficits observed at the functional level, and provide a potential mechanistic link between *XPO7* LoF and impaired neuronal excitability and network synchrony. $Na_v1.2$ is predominantly enriched at glutamatergic synapses across brain regions, including the cortex, hippocampus, olfactory bulb, and striatum [41]. Rodent models and iPSC-derived excitatory neurons, deficient in $Na_v1.2$, consistently exhibit reduced synapse density and impaired synaptic transmission, underscoring its essential role in maintaining synaptic integrity [42, 43]. Disruption of the subcellular distribution of $Na_v1.2$ impairs the delicate balance of Na^+ channels at synapses, which is essential for coordinated neuronal communication, leading to irregular neuronal and network activity. This conclusion is supported by both our imaging-based quantification of synaptic $Na_v1.2$ expression, performed in combination with pre- and postsynaptic markers, and by proteomic analyses of biochemically isolated synaptic fractions. However, direct detection of synaptic $Na_v1.2$ by immunoblotting was not successful, as the $Na_v1.2$ immunoblotting procedure was incompatible with the synapse isolation protocol. While $Na_v1.7$ appeared upregulated in the synaptic proteome, we were unable to validate this finding by immunostaining due to the lack of specific antibodies. We therefore limited our conclusions to $Na_v1.2$. Future studies will aim to determine whether similar sodium channel abnormalities are present in *XPO7* LoF patient-derived cell lines and corresponding mouse models. In this study, we utilized isogenic hiPSC-derived neuronal lines generated from a single donor background to dissect the causal effects of *XPO7* LoF on neuronal and network phenotypes. This experimental design controls for genetic variability and addresses the causal role of *XPO7* disruption; however, it also limits the generalizability of our findings, as inter-individual variation inherent to the human population is not represented. Future work using iPSC lines derived from patients carrying *XPO7* LoF mutations will be able to address the impact of genetic background and generalizability. In parallel, studies employing *XPO7* mutant mouse models will enable evaluation of neuronal and network phenotypes within intact circuits and developmental trajectories that are difficult to model in induced human neurons. Notably, a recent study demonstrated that *XPO7* haploinsufficiency in mice disrupts nuclear transport, impairs neuronal morphogenesis, and leads to cognitive and social behavior deficits that emerge in early adulthood, mirroring the onset period of schizophrenia in humans [15]. Our report, together with recent in vivo studies, establishes an important causal link between *XPO7* loss and schizophrenia-relevant phenotypes.

METHODS

Cell lines

CRISPR-Cas9-mediated homozygous and heterozygous knockout clones of *XPO7* were generated in PGP1 iPSCs, originally derived from primary fibroblasts of the Personal Genome Project donor PGP1 (Dr. George Church, Harvard Medical School) and supplied by EditCo Bio, Inc. (Redwood City, CA, USA). The parental PGP1 iPSCs were authenticated by short tandem repeat (STR) profiling at passage 40 and confirmed to exhibit a normal karyotype and genomic stability. To generate the knockout clones, ribonucleoprotein complexes consisting of Cas9 protein and synthetic, chemically modified guide RNA (ACAGAUUCUCUA-GUUGGGCC) were electroporated into cells. Editing efficiency was assessed 48 hours post-electroporation. Monoclonal populations were established by seeding single cells into 96-well plates using a single-cell printer (Molecular Devices), followed by imaging every 3 days to monitor clonal expansion. Independent clones were genotyped to confirm heterozygous and homozygous editing using PCR with the following primers: forward (5′-3′) TCCTCAAGGGAATGATGAAAAAGT and reverse (5′-3′) ATAACGTTTCC-GAAGCGCCT. Confirmed edited clones were further validated by karyotype analysis and assessment of pluripotency prior to neuronal differentiation. In addition, the genomic integrity of the edited clones was evaluated by SNP microarray analysis. Parental control clones were generated by subjecting cells to the same CRISPR process using a non-targeting guide RNA, followed by identical selection procedures. For the initial characterization of genotypic differences, we used one parental control clone (non-targeting gRNA), one heterozygous *XPO7* knockout clone, and one homozygous *XPO7* knockout clone to model *XPO7* loss of function. To validate key findings, we analyzed an additional clone for both the heterozygous and homozygous *XPO7* knockouts across all major assays, including the voltage-ramp electrophysiological protocol (Fig. S2), HD-MEA (Fig. S4) and LD-MEA experiments (Fig. S7), bulk RNA-seq analyses at days 28, 42, and 56 (Fig. S5), biochemical validation of cytoplasmic and synaptic proteomic data (Fig. S6), and measurements of synaptic density (Fig. S9). A comprehensive summary of the number of clones, differentiation batches, and samples included in each experiment, along with the corresponding figure references, is provided in the Supplementary Information (Supplementary Table S3). All iPSC cultures were routinely screened for mycoplasma contamination using the MycoAlert Mycoplasma Detection Kit (Lonza, LT07-710) and the Universal Mycoplasma Detection Kit (ATCC, 30-1012K) and were confirmed negative.

Induced neuron differentiation

Induced neurons were generated following established protocols with minor modifications [27]. A doxycycline-inducible NGN2 expression construct was introduced into human iPSCs via TALEN-mediated integration at the AAVS1 safe-harbor locus. Briefly, stem cells were cultured in Induction Medium composed of DMEM/F12, Glutamax (1:100), 20% glucose (1.5% v/v), N2 Supplement (1:100), LDN-193189 (200 nM), SB431542 (10 μM), XAV939 (2 μM), and doxycycline (2 μg/mL) to activate NGN2 expression. After 24 hours, the Induction Medium was refreshed to include zeocin (4 μg/mL) for selection. On Day 4, the medium was replaced with Neuron Medium consisting of Neurobasal Medium, 20% glucose (1.5% v/v), Glutamax (1:100), MEM-NEAA (1:100), B27 Supplement (1:50), brain-derived neurotrophic factor (BDNF, 10 ng/mL), glial cell line-derived neurotrophic factor (GDNF, 10 ng/mL), and ciliary neurotrophic factor (CNTF, 10 ng/mL), supplemented with FUDR (10 μM) to inhibit the proliferation of non-neuronal cells. On Day 5, cells were dissociated using Accutase and replated onto Geltrex-coated plates at a density of 40,000 cells/cm^2. Neurons were maintained in Neuron Medium with media changes every three days until downstream assays. For all experiments in this study, except for proteomic and biochemical analyses, induced neurons were co-cultured with primary rat glia to promote maturation and synaptic connectivity.

Automated patch-clamp experiments

Electrophysiological properties of human neurons were recorded using the SyncroPatch 384PE system (Nanion Technologies) with whole-cell patch clamp and gigaohm (GΩ) seals, as previously described [29]. Neurons at day 42 were dissociated using the Papain Dissociation System (Worthington Biochemical). Papain was reconstituted in EBSS, incubated at 37°C until clear, and combined with DNase I (final concentration: 20 units/mL papain, 0.005% DNase). Neurons were incubated in the papain-DNase solution at 37°C for 75 minutes without agitation to prevent curling of detached cell sheets. Following digestion, cells were gently triturated until a mostly single-cell suspension was achieved. The suspension was layered onto an albumin-ovomucoid inhibitor gradient and centrifuged at 1000 rpm for 4 minutes. The resulting pellet was resuspended in a 1:1 mixture of standard external recording solution and Neuron Medium, then gently triturated to ensure a homogeneous suspension. Cell concentration was adjusted to 200,000 cells/mL and maintained at 10°C in a shaking cell hotel prior to loading onto the SyncroPatch platform.

All recording solutions were freshly prepared using ultrapure Milli-Q water (18 MΩ cm), filtered through a 0.22 μm PES membrane. For voltage-dependent activation assays, we used a modified KF110 internal solution

from Nanion, composed of the following components: 110 mM KF, 10 mM KCl, 10 mM NaCl, 1.5 mM $MgCl_2$, 10 mM EGTA, 10 mM HEPES, and 2 mM freshly added Na-ATP, adjusted to pH 7.2 with KOH. For both voltage- and frequency-dependent activation assays, the standard CsF110 internal solution from Nanion was used, composed of the following components: 110 mM CsF, 10 mM CsCl, 10 mM NaCl, 10 mM EGTA, 10 mM HEPES, and 2 mM freshly added Na-ATP, adjusted to pH 7.2 with CsOH. The Standard ECS consisted of 140 mM NaCl, 4 mM KCl, 1 mM $MgCl_2$, 10 mM HEPES, 5 mM glucose, and 2 mM freshly added $CaCl_2$, adjusted to pH 7.4 with NaOH. The Seal Enhancer included 130 mM NaCl, 4 mM KCl, 1 mM $MgCl_2$, 10 mM HEPES, and 10 mM freshly added $CaCl_2$, adjusted to pH 7.4 with NaOH. The Reference Solution was composed of 140 mM NaCl, 4 mM KCl, 10 mM HEPES, 1 mM $MgCl_2$, and 2 mM freshly added $CaCl_2$, adjusted to pH 7.4 with NaOH. All electrophysiological measurements were consistently reproduced across 3 to 7 batches of automated patch-clamp experiments and validated using neurons from two independent differentiation batches (Fig. S2).

Manual patch-clamp recording

Coverslips containing induced neurons were placed in the recording chamber on the stage of an inverted microscope (Axiovert.A1, Zeiss, Oberkochen, Germany) equipped with phase-contrast optics. All recordings were conducted at room temperature (24°C-26°C). Recording electrodes were fabricated using a horizontal P-1000 pipette puller (Sutter Instrument, Novato, CA, USA). Membrane potentials were recorded in the current-clamp whole-cell configuration using pipettes filled with a K-gluconate-based internal recording solution. The same internal solution was also used for voltage-ramp recordings. Whole-cell voltage-clamp recordings were performed at a holding potential of -70 mV using a MultiClamp 700B Amplifier (Molecular Devices, Sunnyvale, CA, USA) in gap-free mode for at least 5 minutes per cell. Access resistance was continuously monitored throughout the recordings. Data were digitized at 10 kHz and filtered with a 2-kHz low-pass filter. The liquid junction potential, calculated as 14.8 mV at 25°C, was not corrected. Data were analyzed offline using Clampfit 10.02 (Molecular Devices) and the MiniAnalysis Program (Synaptosoft, Fort Lee, NJ, USA).

All recording solutions were prepared with ultrapure Milli-Q water (18 MΩ cm), filtered with 0.22 μm PES membrane, and stored at 4°C until use. K-gluconate-based internal recording solution: 131 mM K-gluconate, 17.5 mM KCl, 1 mM $MgCl_2$, 10 mM HEPES, 1 mM EGTA, 2 mM Mg-ATP, and 0.2 mM Na-GTP, adjusted to pH 7.4 with KOH. Extracellular solution: 145 mM NaCl, 5 mM KCl, 1 mM $MgCl_2$, 2 mM $CaCl_2$, 5 mM HEPES and 5 mM glucose, adjusted to pH 7.4 with NaOH.

High-density microelectrode arrays

The HD-MEAs were prepared according to the manufacturer's protocols with minor modifications. Plates were sterilized with 70% ethanol, rinsed with sterile deionized water, and pre-conditioned with complete culture medium for two days at 37°C in 5% CO_2. Surfaces were coated with Poly-D-Lysine (0.1 mg/mL) for 3 hours, followed by a Geltrex (1:100) coating for 1 hour at 37°C. Induced neurons were plated on Day 5 at a density of 40,000 cells/cm^2 onto the electrode array and incubated for 1 hour to allow attachment. Pre-warmed Neuron Medium was added after incubation, and cells were maintained at 37°C in 5% CO_2 and > 95% humidity, with half-medium changes performed three times per week. Neuronal activity was recorded 16-24 hours after each medium change to ensure consistent timing between media replacement and recording sessions.

HD-MEA recordings were performed using the MaxTwo HD-MEA system (MaxWell Biosystems) in accordance with the manufacturer's instructions. The recording setup was maintained inside a 5% CO_2 cell culture incubator at 37°C. Recordings were conducted using the Activity Scan Assay, Network Assay, and AxonTracking Assay modules within the MaxLab Live software (MaxWell Biosystems), following established protocols. Spontaneous neuronal activity across the array was recorded using 6600 electrodes with a 35 μm pitch across seven electrode configurations for 120 seconds. Subsequently, the 1024 most active electrodes, identified based on firing rates, were selected to record network electrical activity for 900 seconds. Active electrodes were determined by their firing frequency, and those with the highest activity were prioritized for extended recordings.

HD-MEA data analysis

Our data processing pipeline for extracting single-neuron electrophysiological features followed established protocols [31, 33]. Raw recording data were filtered and spike-sorted using Kilosort2 within the SpikeInterface framework [58], employing default parameters to identify individual neuronal units. We then automatically curated the spike sorting output using parameters adapted from a previous study [33]. An inter-spike interval violation threshold≤ 0.5 was used to assess refractory period violations, with lower values indicating reduced contamination from overlapping spike trains. A firing rate threshold≥ 0.05 Hz excluded units with minimal or inconsistent activity. A signal-to-noise ratio threshold≥ 3 ensured that each unit's waveform was clearly distinguishable from background noise. An amplitude cutoff≤ 0.1 estimated the proportion of potentially missed spikes based on deviations from a smoothly declining amplitude distribution. A nearest-neighbours hit rate≥ 0.8 verified that most of a unit's nearest neighbours in principal component space belonged to the same cluster, indicating good unit isolation. For the extraction of network metrics (Fig. 6), we additionally computed synchrony metrics using SpikeInterface, which quantify the occurrence of coincident spikes across multiple spike trains, applying a synchrony threshold of 0.01. This approach ensured that network analyses were restricted to units actively participating in synchronous spiking events.

Single-cell features were subsequently extracted from these curated spike-sorted units using either SpikeInterface or Deephys [32], an open-source tool designed for functional phenotyping of in vitro neuronal cultures recorded with high-density microelectrode arrays. The extracted features included activity metrics, waveform metrics, and propagation metrics. Activity metrics comprised the median spike amplitude, amplitude coefficient of variation, mean interspike interval, interspike interval coefficient of variation, and firing rate. Waveform metrics included peak-to-valley duration, defined as the time between the negative and positive peaks; half-width duration, defined as the signal duration at 50% of the spike amplitude; peak-to-trough ratio, the ratio of negative to positive peak amplitudes; recovery slope, the rate of return from the negative peak to baseline; and repolarization slope, the rate of return from the positive peak to baseline. Propagation metrics, derived by tracking action potential propagation from the axon initial segments, included conduction velocity, maximum axonal length, and additional measures that characterize the dynamics of action potential propagation in individual neurons.

Low-density microelectrode arrays

Induced neurons were also recorded using low-density multielectrode arrays (Multichannel Systems, MCS GmbH, Reutlingen, Germany). Each well contained 9 electrodes, each with a diameter of 30 μm, spaced 300 μm apart. Neuronal network activity was recorded for 10 minutes after a 10-minute acclimatization period in a recording chamber maintained at 37°C with 5% CO_2. Raw signals were sampled at 10 kHz and processed using a high-pass second-order Butterworth filter with a 100 Hz cutoff frequency and a low-pass fourth-order Butterworth filter with a 3500 Hz cutoff frequency. Spike detection was performed by setting the noise threshold at ± 4.5 standard deviations to accurately capture neuronal firing events.

RNA sequencing

RNA extraction, library preparation, and sequencing were performed at Azenta Life Sciences (South Plainfield, NJ, USA). Total RNA was extracted from frozen cell pellets using the Qiagen RNeasy Plus Micro Kit according to the manufacturer's protocol. RNA concentration was measured using the Qubit 2.0 Fluorometer (Life Technologies), and RNA integrity was assessed with the Agilent TapeStation 4200 (Agilent Technologies). RNA sequencing libraries were prepared using the NEBNext Ultra II RNA Library Prep Kit (NEB) with poly(A) selection. mRNA was enriched using Oligod(T) beads, followed by fragmentation at 94°C for 15 minutes. First and second-strand cDNA synthesis was performed, followed by end-repair, adenylation at the 3′ ends, and ligation of universal adapters. Indexed libraries were enriched by PCR, validated using the Agilent TapeStation, and quantified using both the Qubit 2.0 Fluorometer and quantitative PCR (KAPA Biosystems). The libraries were clustered on a flow cell and sequenced on an Illumina NovaSeq platform using a 2 × 150 bp paired-end configuration. Image analysis and base calling were conducted using Illumina's Control Software, and raw data (.bcl files) were converted into FASTQ files and de-multiplexed using bcl2fastq 2.17, allowing for one mismatch in the index sequence.

RNA sequencing data analysis

Raw reads were trimmed to remove adapter sequences and low-quality bases. The trimmed reads were aligned to the ENSEMBL reference genome

using the STAR aligner (v.2.5.2b), generating BAM files. Unique gene hit counts were obtained using featureCounts from the Subread package (v.1.5.2), counting only unique reads within exon regions. Differential gene expression analysis was conducted using DESeq2, applying the Wald test to calculate p-values and $\log_2$ fold-changes. Genes with adjusted p-values < 0.01 and absolute $\log_2$ fold-changes > 1 were considered differentially expressed.

Gene set enrichment analysis

GSEA was conducted using the clusterProfiler package from R Bioconductor [39], incorporating the C5: GO gene sets from the Molecular Signatures Database (MSigDB) [37], the SynGO collection [39] and gene sets curated from the literature [40]. For proteomics data, proteins were pre-ranked according to their moderated $\log_2$ fold-change for each comparison (*XPO7*$^{+/-}$ vs. parental control or *XPO7*$^{-/-}$ vs. parental control). When multiple isoforms were present, the isoform with the largest effect size (absolute $\log_2$ fold-change) was selected for pre-ranking. Gene Ontology (GO) terms with a false discovery rate (FDR) < 0.05 were considered significant. The GO terms highlighted in the main figures represent the top-ranking categories with the greatest statistical significance in either *XPO7*$^{+/-}$ or *XPO7*$^{-/-}$ cells. All results across datasets are provided in the Supplementary Tables.

Immunofluorescence and image analysis

Immunofluorescence was performed on day 42. Induced neurons were rinsed with ice-cold PBS and fixed with 4% paraformaldehyde (PFA) supplemented with 4% sucrose for 15 minutes at room temperature. Following fixation, cells were washed with PBS and permeabilized using PBS containing 0.2% Triton X-100 for 10 minutes. Samples were then blocked with BlockAid blocking solution for 30 minutes at room temperature. Primary antibodies, diluted in BlockAid, were applied and incubated overnight at 4°C. The following antibodies were used in this study: MAP2 (1:2000, Abcam, ab5392), PSD-95 (1:100, BioLegend, 810401), Synapsin I/II (1:1000, Synaptic Systems, 106004), Anti-Sodium Channel (Pan-Na_v, 1:1000, Sigma-Aldrich, S8809), $Na_v1.2$ (1:200, Alomone Labs, ASC-002). After primary antibody incubation, cells were washed with PBST and incubated with secondary antibodies, also diluted in BlockAid, for 1 hour and 30 minutes at room temperature. The secondary antibodies used included goat anti-chicken 488 (1:2000), goat anti-chicken 405 (1:250), goat anti-mouse 555 (1:1000), and goat anti-guinea pig 647 (1:1000). When applicable, DAPI (5 μg/mL) was applied for 10 minutes at room temperature for nuclear staining. Stained cells were imaged using the Opera Phenix High-Content Screening System (PerkinElmer). A minimum of 25 fields with 5 z-stacks per well were acquired using either 20 × or 63 × water immersion objectives in confocal mode. Microscopy images were exported and analyzed using Signals Image Artist (Revvity) for quantitative assessment and visualization. To ensure uniformity, the analysis was performed using a batch imaging analysis system so that all wells within the same plate, containing different genotypes, were processed simultaneously under identical parameters. Image processing incorporated several functional modules, including cell body recognition, neurite tracing, synaptic puncta identification, and fluorescence intensity quantification, all implemented through pre-established algorithms within the batch analysis platform (Signals Image Artist, Revvity Inc). Neuronal soma were first identified based on DAPI and MAP2 staining, followed by tracing of neurites and their branches extending from each soma. Synapse detection was restricted to regions in close proximity to dendritic shafts. Synaptic puncta were defined as regions positive for both Synapsin I/II and PSD-95, markers of presynaptic vesicles and postsynaptic densities, respectively. $Na_v1.2$ expression at synapses was quantified by measuring the $Na_v1.2$ fluorescence intensity within regions defined by the colocalization of Synapsin I/II and PSD-95. The same workflow and parameters were applied consistently to all images and wells during batch processing.

Isolation of Synaptosomes

The isolation of synaptosomes was performed following a previously described protocol [46]. Briefly, day 42 neurons cultured in 6-well plates without rat glia coculture were washed twice with ice-cold PBS, followed by the addition of 1 mL Buffer A (10 mM HEPES, pH 7.4, 2 mM EDTA, 5 mM sodium orthovanadate, 30 mM sodium fluoride, 20 mM β-glycerol phosphate, and 1 tablet of protease inhibitor cocktail per 50 mL) to each well. Cells were scraped, collected, and transferred to a glass douncer for homogenization (30 strokes). The homogenate was centrifuged at 500 × g for 5 minutes at 4°C to pellet cell debris, nuclei, and extracellular matrix. The resulting supernatant was then centrifuged at 10,000 × g for 15 minutes at 4°C to isolate the synaptosomal fraction. The crude synaptosomal pellet was resuspended in 200 μL of Buffer B (10 mM HEPES, pH 7.4, 2 mM EDTA, 2 mM EGTA, 5 mM sodium orthovanadate, 30 mM sodium fluoride, 20 mM β-glycerol phosphate, 1% Triton X-100, and 1 tablet of protease inhibitor cocktail per 50 mL) for downstream purification, immunoblotting, or mass spectrometry. Throughout the procedure, care was taken to avoid disturbing soft pellets, and all buffers were pre-chilled and supplemented with protease inhibitors to maintain sample integrity.

Mass spectrometry

Tryptic peptides were dried in a vacuum centrifuge and resuspended in 100 μL of 100 mM triethylammonium bicarbonate (TEAB) buffer, followed by vortexing and brief centrifugation. TMT label reagents were equilibrated to room temperature, dissolved in 20 μL of anhydrous acetonitrile, vortexed, and briefly centrifuged. After 5 minutes, 20 μL of the TMT reagent was added to each 100 μL peptide sample, vortexed, centrifuged, and incubated for 1 hour at room temperature. The reaction was quenched with 5 μL of 5% hydroxylamine and incubated for 15 minutes. Labeled samples were pooled in equal amounts and dried using a speed-vac. Fractionation was performed using the Pierce High pH Reversed-Phase Peptide Fractionation Kit following the manufacturer's protocol. Dried fractions were resuspended in 0.2% formic acid before LC-MS analysis. LC-MS Analysis was conducted using a Thermo Ultimate 3000 HPLC coupled to an Orbitrap Exploris 480 mass spectrometer. TMT-labeled peptides were separated on a Thermo PepMap RSLC C18 column (2 μm, 75 μm × 50 cm) using a multi-step gradient of solvent A (0.1% formic acid in water) and solvent B (0.1% formic acid in acetonitrile) at a flow rate of 200-300 nL/min. The mass spectrometer operated in data-dependent mode with full MS scans at 60,000 resolution across 450-1600 m/z, followed by MS/MS scans (NCE 32) at 45,000 resolution with a 30 s dynamic exclusion and a 3 s cycle time.

Proteomic data analysis

Raw mass spectral data files (.raw) were analyzed using Sequest HT within Proteome Discoverer (Thermo Scientific). The search parameters were set as follows: a precursor ion mass tolerance of 10 ppm and a fragment ion mass tolerance of 0.05 Da. Up to two missed trypsin cleavages were allowed. Fixed modifications included carbamidomethylation of cysteine residues and TMT modifications on lysine residues and peptide N-termini. Variable modifications encompassed methionine oxidation, methionine loss at the protein N-terminus, N-terminal acetylation, and methionine loss combined with N-terminal acetylation. Data were searched against the UniProt Human database (UP000005640) and an in-house generated contaminant database to ensure accurate identification and exclusion of non-specific hits. Following median normalization of the datasets, a moderated two-sample t-test was applied to compare the *XPO7*$^{+/-}$ group with the parental control and the *XPO7*$^{-/-}$ group with the parental control, respectively. This statistical approach enabled robust detection of differentially expressed proteins while effectively accounting for variance across samples. To validate these results, key differentially expressed proteins from both datasets were examined using western blotting in two independent clones for both *XPO7*$^{+/-}$ and *XPO7*$^{-/-}$ cells. The following antibodies were utilized for western blot validation: NPTX2 (1:2000, Proteintech, 10090-910), PSD95 (1:1000, Cell Signaling Technology, 3450S), Synaptophysin (1:1000, Cell Signaling Technology, 36406S), Synapsin3 (1:1000, Synaptic Systems, 106303), VGLUT2 (1:1000, Cell Signaling Technology, 71555S), XPO7 (1:1000, Novus Biologicals, NBP3-35644), Vinculin (1:200, Sigma-Aldrich, V9131), NECAB2 (1:1000, Sigma-Aldrich, HPA013998), CD166 (1:1000, Abcam, ab109215), ADAM11 (1:1000, Neuromab, N441/35), Plexin A4 (1:1000, Cell Signaling Technology, 3816S).

Statistical analysis

Sample, clone, and batch sizes for each genotype and assay are detailed in the Supplementary Information (Table 1). Sample sizes were determined a priori based on pilot electrophysiological recordings obtained from a single clone per genotype, which provided variance estimates used to ensure sufficient power to detect the expected genotype-dependent differences in Na^+ channel current density and neuronal excitability at the level of independent differentiation batches. To confirm the robustness of these findings, the key assays were subsequently validated using an additional independent clone for each genotype.

Automated platforms were used for patch-clamp, MEA, and imaging assays, with *XPO7*$^{+/+}$, *XPO7*$^{+/-}$, and *XPO7*$^{-/-}$ lines seeded on the same assay plates and processed in parallel within each differentiation batch. For every differentiation, at least two assay plates containing all genotypes were prepared, and the seeding positions of individual genotypes were rotated across plates to minimize potential positional effects. Data acquisition and analysis were performed concurrently using automated pipelines under identical parameters. For experiments not involving automated pipelines, including manual patch-clamp recordings, RNA-seq sample preparation, and proteomic processing, data collection and sample handling were performed under blinded conditions. All samples that met standard quality control criteria were included in the analyses, and no data were excluded.

Statistical analyses were performed using GraphPad Prism 8 (GraphPad Software, Inc., CA, USA). For experiments conducted at a single age, comparisons among multiple groups were performed using one-way ANOVA followed by Dunnett's post hoc test. For analyses involving multiple age points, two-way ANOVA was used to assess the effects of age and genotype, followed by Dunnett's post hoc test for multiple comparisons. In addition to conventional ANOVA-based analyses, linear mixed-effects models were applied to verify significant effects while accounting for random variability across differentiation batches and clones. Both approaches yielded consistent results. In cases where only two groups were compared at a single time point, unpaired Student's t-tests were used. Results with p-values less than 0.05 were considered statistically significant and are denoted as follows: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***) and $p < 0.0001$ (****). Variation within each group is represented by the standard error of the mean (SEM). Each data point corresponds to an independent biological replicate, with the median of technical replicates calculated prior to statistical analysis. For HD-MEA and high-content imaging experiments, multiple cells within each well were analyzed, and the median value of each measurement across all cells in a well was used to represent that well as a single data point for both visualization and statistical testing. Data distributions were assessed for normality. When deviations from normality were detected, such as in highly skewed MEA datasets with many zero values at early developmental stages, Box-Cox transformations were applied to approximate a normal distribution prior to conducting parametric statistical analyses [59].

DATA AVAILABILITY

Raw bulk RNA sequencing FASTQ files have been deposited in the NCBI Gene Expression Omnibus (GEO) database (GSE301134). All datasets are publicly available. This study does not report any original code. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

J.Q.P. conceived the study and supervised the project. J.Q.P. and L.C. designed the methodology. J.Q.P., L.C., E.K., D.H., R.L. and Y.W. performed the investigation. J.Q.P. and L.C. wrote the original draft. J.Q.P., R.N., M.S., A.H., P.H. and L.C. reviewed and edited the manuscript. J.Q.P. acquired funding and, together with R.N., provided resources. All authors reviewed and approved the manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

All methods were performed in accordance with relevant guidelines and regulations. This study did not involve human participants or live vertebrate animals. Human induced pluripotent stem cells used in this study were obtained from commercial sources. The providers of these cell lines certify that all samples were collected with appropriate ethical approval and informed consent from donors, in compliance with applicable regulations. Therefore, approval from an ethics committee was not required for this study. Consent to participate is not applicable, as no human participants were directly involved in this research.

ADDITIONAL INFORMATION

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